# Dividing the Costs of War: Conscript Fatalities and Demands for Progressive Taxation in Interwar Britain*

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**Abstract**

The compensatory theory of tax fairness claims that governments implemented progressive taxation schemes during and after the World Wars to address concerns that wartime conscription had unfairly imposed the burden of fighting on the poor. I offer a new test of the compensatory theory that examines whether subnational variation in World War I conscript fatalities is associated with support for progressive taxation. Using a geocoded dataset of the home addresses of 446,295 British war dead, I find that constituencies with higher conscript fatality rates lent greater support to left-wing political candidates. I also show that members of Parliament representing high fatality constituencies voted in favor of progressive taxation more frequently, with electorally vulnerable Conservative MPs exhibiting the largest shifts in voting behavior. These findings provide substantial new evidence that fairness concerns arising from wartime conscription influenced postwar redistributive policy and operated through individual legislators' electoral incentives.

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# 1 Introduction

The first widespread implementation of progressive taxation took place in democracies during and after the World Wars. Governments involved in these conflicts introduced marginal income tax rates as high as 91% and imposed new levies on inheritances, financial transactions, and corporate profits (Cronin, 2005; Piketty and Saez, 2007). Many of the new taxes continued after the wars, and the greater fiscal capacity these schemes enabled has been credited with fueling state expansion in the 20th century, midwifing the creation of the welfare state, and contributing to declines in economic inequality in Europe and North America (Piketty, 2014; Titmuss, 1959). While it is not surprising that the fiscal demands of war induced states to broaden taxation, the new levies' progressiveness and persistence after the wars has long puzzled scholars.

One explanation for the jump in tax progressiveness during the World Wars is the compensatory theory, which posits that demands for taxing the rich grew out of a perception that wartime government policy had unjustly favored the wealthy (Seligman, 1893; Scheve and Stasavage, 2016). During the wars, governments implemented policies that had disproportionate impact on the poor, such as conscripting young men and curtailing the rights of workers in military-critical industries (Daunton, 2002). Proponents of the compensatory theory argue that such policies evoked feelings of unfairness among the affected, who then demanded that those who had escaped conscription or profited off the war be taxed as recompense. Progressive taxation would restore equal treatment by ensuring those who had borne the human costs of war would not also have to bear its financial costs.

Past research has offered convincing evidence for several predictions of the compensatory theory. Scheve and Stasavage (2010, 2012) find that states mandating conscription during the World Wars were more likely to introduce progressive income and inheritance taxes. Importantly, they also demonstrate that UK members of Parliament (MPs) supporting of progressive income taxes invoked compensatory justifications more frequently after the start of World War I (Scheve and Stasavage, 2016). Seeking to validate the compensatory theory's microfoundations, scholars have also employed survey experiments to test whether perceptions of unfairness in government policy affect individuals' tax preferences. While primarily conducted in peacetime contexts, these experiments have shown that individuals express greater support for progressive taxation when presented with examples of government policies that benefit the wealthy (Alvarado, 2024; Alvarado et al., 2025; Zakharov and Chapkovski, 2025).

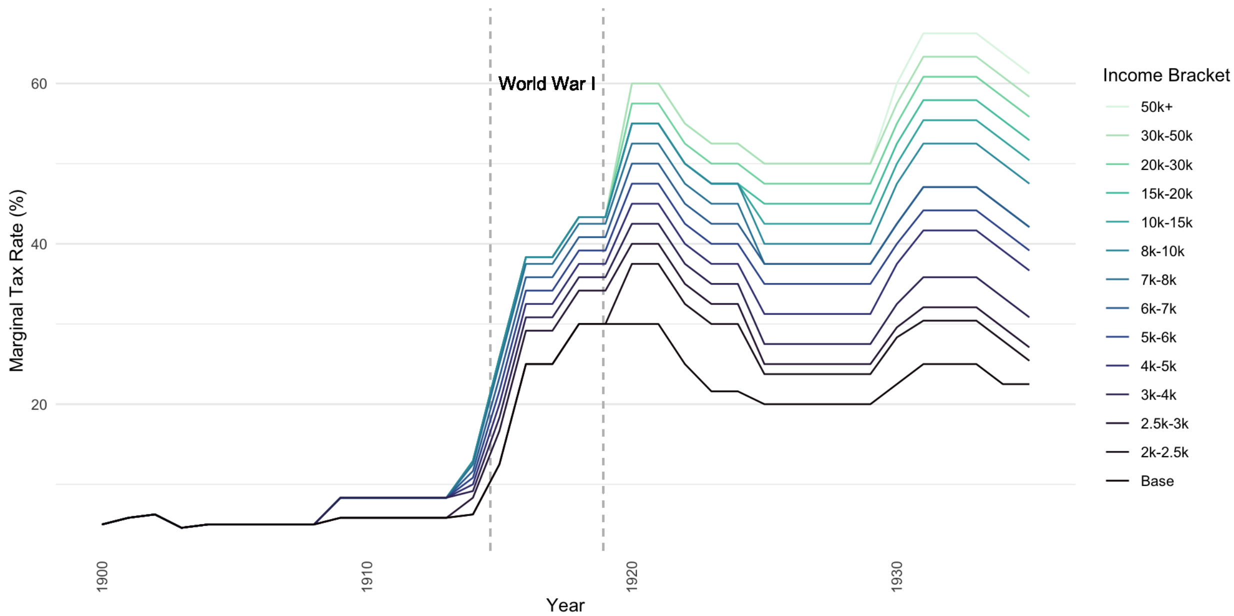


Figure 1: UK income tax rates by bracket, 1900-1935, from *Hansard*

Despite the evidence offered in support of the compensatory theory, several empirical questions remain. First, while existing work has documented the macro (policy) and micro (individual) predictions of the compensatory theory, less attention has been paid to verifying the mechanism that connects individual fairness concerns to national tax policy. Though it has been implied that progressive taxation was implemented by legislators in response to popular demands, direct quantitative evidence documenting whether and how popular pressure was exerted is sparse. Second, little work has examined to what extent compensatory arguments have a persistent effect on taxation after the instigating government policy. As shown in Figure 1, British income tax progressiveness, whether measured by the top marginal tax rate or total number of tax brackets, persisted in the decades after WW1 ended and was expanded multiple times. It remains unclear whether this delayed expansion of progressive taxation was influenced by experiences of conscription during the war. Finally, much of the macro-level quantitative evidence mustered in support of compensatory explanations for wartime tax policy comes from difference-in-differences and event study designs. Using national time series data, these approaches compare tax policy and MP invocations of compensatory arguments before and after conscription was implemented. In these types of designs, interpreting the effect of conscription on progressive taxation as causal requires ruling out the presence of simultaneous confounding events that also influenced tax policy (Miller, 2023; Roth et al., 2023). This is a demanding requirement. The World Wars were transformational events–conscription occurred concomitantly with mass female entry into the workforce, the expansion of the franchise, the accumulation of large public debts, rapid technological advancement, and the mobilization of a war economy (Boehnke and Gay, 2022; Daunton, 2002). Confidence in the predictions of the compensatory theory would be strengthened by within-country cross-sectional variation showing that exposure to unfair wartime government policy influenced the passage of progressive tax policy.

In this research note, I use the case of interwar Britain to present new evidence clarifying the mechanism linking perceptions of unfairness to the implementation of progressive taxation. My approach examines how fatalities of UK soldiers conscripted to fight in World War I affected voter and legislator support for progressive taxation. The deaths of the involuntarily enlisted generated feelings of injustice among the deceased's associates, who observed some wealthy individuals evade conscription or even profit from the war (Daunton, 2002; Harrison, 1971). Conscript fatalities thus serve as a useful local proxy for sentiments of unfair wartime treatment. I estimate the WWI conscript fatality rate for all 509 parliamentary constituencies in England and Wales by combining a geocoded dataset of the home addresses of 446,295 British war dead with over 39 million records from the 1911 UK census. Using historical UK Army organizational records, I show that, conditional on regimental recruitment units, the conscript fatality rate is plausibly exogenous to political behavior and a result of battlefield outcomes. With this measure of local perceptions of unfair government treatment, I carry out a two-part empirical analysis that illustrates an electoral mechanism for the compensatory theory. Specifically, I show that the passage of progressive taxation legislation in interwar Britain was driven at least partially by electoral pressure on Conservatives arising from voter support for left-wing political parties in areas with high conscript fatalities.

First, I demonstrate that candidates representing the Labour Party and other economically left-wing political parties received larger vote shares in constituencies with greater WWI conscript fatality rates. In general elections between 1918 and 1935, left-wing parties increased their share of the vote by an expected 0.37 percentage points for each percentage point increase in the fatality rate of a constituency's conscripts. Counterfactual election simulations reveal that war fatalities had important implications for control of Parliament. A one-standard deviation increase in the conscript fatality rate (roughly 3%) would have been sufficient to flip an additional 87 seats to left-wing candidates during the period under study. As economically left-wing parties advocated for progressive taxation and other redistributive measures, the fact that they received greater electoral in high-fatality constituencies indicates that conscript fatalities shifted voter preferences toward progressive taxation.

Second, using a new dataset of 424 roll-call votes on tax legislation debated in the House of Commons between 1918 and 1935, I show that MPs representing high-fatality constituencies were more likely to vote in support of progressive taxation legislation, regardless of party affiliation. The effect of conscript fatalities on MP support for progressive taxation was strongest for members of the Conservative Party, whose traditional opposition to progressive taxation became an electoral vulnerability. A one-standard deviation rise in the conscript fatality increased the frequency of pro-progressive taxation votes by 13% among Conservative MPs. I also find that conscript fatalities had stronger effects on MP behavior during Parliaments that followed elections where fatalities provided larger electoral benefits to left-wing political parties. Taken together, the results suggest that progressive taxation was implemented in response to electoral pressure arising from constituent reactions to war fatalities. Furthermore, they show that electoral impact of war fatalities continued to influence policy debates over taxation into the 1930s.

In addition to providing new empirical evidence on the mechanisms of the compensatory theory, this note contributes to a large literature on the domestic political impacts of wartime casualties. Past empirical studies, conducted primarily in the context of the Vietnam War and the US War on Terror, have argued that wartime casualties are associated with reduced political legitimacy and incumbent electoral support but greater political participation (De Mesquita and Siverson, 1995; Gartner and Segura, 1998; Gartner, Segura and Barratt, 2004; Koch and Nicholson, 2016; Mueller, 1973; Stam, 1999). Other related research examines how those affected by war mobilize to demand compensatory government support (Dorr and Shin, 2021; Haffert, 2019; Skocpol, 1995), the relationship between war and inequality (Piketty, 2014; Scheidel, 2018), and the role of fairness in determining public opinion on redistributive social policy (Cavaille, 2023). This note builds on both sets of studies by looking at how war fatalities influence support for specific policies, namely progressive taxation, and offering empirical evidence linking voter-level experiences of war to the passage of taxation legislation.

This note also adds to an emerging research agenda employing micro-level conflict data from the World Wars to study combat motivation and political behavior (Acemoglu et al., 2022; Boehnke and Gay, 2022; Cagé et al., 2023; De Juan et al., 2024; Huff, Min and Schub, 2024; Rozenas, Talibova and Zhukov, 2024). Like Acemoglu et al. (2022), who study the political consequences of WWI fatalities in Italy, my study concludes that greater local fatalities are associated with support for socialist and trade unionist political parties. However, my findings contrast with those of De Juan et al. (2024), who show that German WWI fatalities are associated with support for right-wing candidates. While determining the political effects of war casualties across states is outside the scope of this note, my findings, together with previous research, suggest that war-related grievances push voters toward extreme political positions, but the partisan direction of this effect is contingent on other factors, such as whether a state was victorious in the conflict. This paper also contributes to this literature by going beyond electoral outcomes to study the effect of war fatalities on historical government tax policy.

## 2 War fatalities and taxation in Interwar Britain

In this section, I offer historical evidence that perceptions of unfairness in wartime government policies, including conscript deaths, increased demands for compensatory taxation in Britain. These demands manifested as electoral support for left-wing political parties, which ran on platforms advocating for progressive taxation. After the war, MPs observed the shift in voter preferences and responded by casting votes in favor of progressive taxes. From these historical anecdotes, I also derive three empirical hypotheses that use conscript fatalities as a proxy for discontent over unfair wartime policies.

On August 4, 1914, the United Kingdom declared war on the German Empire, thus entering what would become known as World War I. The ensuing four years of fighting created unprecedented logistical, fiscal, and personnel demands on the British state. The government mobilized the economy to ensure a sufficient supply of foodstuff, munitions, and equipment required for the war (Daunton, 2002). The need for soldiers to maintain the stalemate of the Western Front led to the introduction of conscription in Great Britain in 1916, swelling the ranks of the British Army by 5.7 million men, of which over 800,000 would perish in combat (UK War Office, 1922). Financing the expenditure necessary to purchase equipment and pay soldiers led the state to expand taxation and issue bonds, causing public debt to surge from 30% of GDP in 1913 to 144% of GPD in 1918 (Office for Budget Responsibility, 2025).

Though the consequences of Britain's entry into the war were universally felt, there was widespread sentiment that government policies meant to support the war effort disproportionately harmed the poor. Feelings of unfair treatment among the working class were stoked by the passage of laws limiting the rights of workers to ensure production of crucial military supplies was not disrupted. For instance, the Munitions Act of 1915 forbade strikes and limited the ability of workers to leave their jobs (Cronin, 2005). Restrictions on workers' rights were seen as particularly galling in light of the profits earned by the owners of firms supplying ordnance to the government. The introduction of conscription generated additional discontent that the government was expending the bodies of the poor while paying off the owners of armament firms.

Such sentiments were exemplified by the words of Dockers' Union leader Ben Tillett: "the capital classes were sitting at home in comfort and security behind the bodies of men better than themselves.[1]"

Conscription galvanized the nascent trade unionist movement. The Labour Party alleged that state power was being used to the benefit of capital, generating calls for a "conscription of riches" to offset the conscription of bodies. The idea of the conscription of riches crystallized into the policy of a capital levy, a one-time tax on wealth (Daunton, 2002). While the wartime coalition government resisted such a levy, some concessions to the growing labor movement were made. For example, the government implemented the Excess Profits Duty in 1915, an 50% tax on corporate profits that exceeded a prewar standard, intended to prevent "excessive returns to capital" (Billings and Oats, 2014; Cronin, 2005). Yet these concessions did not halt the growth of the labor movement, as evinced by the fact that number of workers going on strike expanded from 300,000 in 1914 to 2 million in 1919 (Cronin, 2005). The Labour Party emerged from the war with a stronger electoral profile, catalyzing its transformation from a trade union lobbying group to a political party with a goal of national governance (Douglas, 1972). This suggests the first empirical hypothesis. If the experience of conscript fatalities increased the appeal of compensatory taxation for voters, left-wing parties that championed such policies should have received greater vote shares in constituencies that lost more of their conscripts during the war.

*H1: Economically left-wing political parties received more electoral support in constituencies with greater conscript fatalities.*

The vitalization of left-wing politics in response to wartime conscription swayed postwar fiscal policy. The United Kingdom emerged from the war with a public debt of £6.1 billion pounds, an almost tenfold increase from 1914 (Office for Budget Responsibility, 2025). Prime Minister David Lloyd George's Conservative-Liberal Coalition government, which had won reelection in 1918, was therefore faced with the need to increase taxes immediately after the end of the war. Despite a consensus on rapid repayment of war debt, there was significant debate over how

[1]Quoted in Harrison (1971).

the burden of debt payment should be divided among social classes. Parliamentary leaders were aware of the discontent over conscription among workers, which was compounded by a decline in living standards due to an economic recession that reduced British economic output by a quarter between 1918 and 1922, feeding resentment over perceived unjust wealth accumulation (Daunton, 2002).

The atmosphere of working class discontent made the prospect of raising revenue through the traditional Conservative method of indirect taxation electorally dangerous. Doing so might drive voters to Labour, which had campaigned on platform of taxing the wealthy though highly progressive income taxes and a capital levy (Daunton, 1996). The threat of electoral backlash prompted the Coalition government to pursue a progressive postwar tax policy, even over the protests of both party members and financiers. In 1920, the income tax was reformed, increasing the number of income tax brackets to 11 and lifting the highest marginal tax rate to 60% on incomes over £50,000, while keeping the base tax rate constant. Moreover, a Conservative Chancellor proposed a war wealth levy in 1920 on additional wealth generated during the war, the nearest a capital levy came to being implemented. The justification for these policies was to "contain Labour", as illustrated by Chamberlain's rebuff to Conservative MPs opposed to the expansion of the Excess Profits Duty: "It is good for them to know that I stand or fall by my proposals. If they won't take them from me, they will get them and a Capital Levy from someone else," referencing the consequences of a Labour election victory.[2] From the postwar push by MPs to expand progressive taxation I postulate a second hypothesis. MPs, assuming they sought to remain in office and were aware of voter preferences, should respond to shifts in voter preferences by casting more votes in favor of progressive taxation. If H1 is true, voter preferences for progressive taxation would be more widespread in constituencies with higher war fatalities, incentivizing MPs to support progressive taxation to avoid losing office.

*H2: Members of Parliament representing constituencies with greater conscript fatalities voted in favor of progressive taxation more frequently.*

[2]Quoted in Daunton (1996).

Yet some Coalition MPs feared that the Coalition did not go far enough in addressing popular concerns about the just distribution of taxes. For example, Winston Churchill, argued that if the Coalition failed to institute a war wealth levy:

> It will be said that we are in the grip of the plutocracy and it will be said with a certain truth. It will be very hard anyway to hold this immense electorate by reason and not by force and still hold the capitalist system. If we cannot reason with them and convince them, we shall bring the very disaster which the City fears.[3]

The statements of Chamberlain and Churchill indicate that electoral pressure to support progressive taxation should be most acute for Conservative MPs. Labour and other left-wing parties already supported progressive taxation for ideological reasons, and therefore would not be subjected to additional electoral pressure. In contrast, if H1 holds, Conservatives MPs in high-fatality districts would be at risk of losing their seats if they disregarded voter demands. This leads to the final hypothesis:

*H3: The effect of conscript fatalities on support for progressive taxation was most significant for members of the Conservative Party.*

In the Online Appendix, I validate the logic of the hypotheses using a game theoretic model of electoral competition over the distribution of public debt payments between social classes. I now devise and carry out a systematic empirical test of these three hypotheses.

# 3 Data

My empirical analysis assesses whether World War I fatalities affected left-wing candidate vote share and MP voting behavior in England and Wales between 1918 and 1935. I focus on England and Wales for reasons of data availability and comparability. In Ireland, conscription was never applied due to concerns about political instability, meaning all Irish soldiers who participated in the war were volunteers (Ward, 1974). Though many Irish soldiers perished during the war, the

[3] Quoted in Daunton (2023).

fact that their sacrifice was voluntary likely means that their deaths produced different reactions than in Great Britain, where most soldiers were unwilling conscripts. Scotland is not included due to inaccessible microdata for the 1911 census, preventing construction of the independent variable.

### 3.1 World War I conscript fatality rates

The independent variable of this study is the combat fatality rate of conscripts from a given parliamentary constituency, which is defined as

$$\text{Conscript Fatality Rate} = \frac{\text{Number of World War I Fatalities}}{\text{Number of Conscripts}} \times 100$$

To determine the number of deceased WWI soldiers who resided in each constituency, I geocoded records of war dead scraped from the website of the Commonwealth War Graves Commission (CWGC), an organization dedicated to preserving the memory of soldiers from the Commonwealth of Nations who died during the World Wars. Each fatality record contains the name, age, burial location, and service record of the fallen soldier, often along with a short epitaph. These epitaphs generally list the name and address of the closest relatives of the deceased, often a spouse and/or parents. The listed addresses can be precise to the house number, and are current at time of burial for the deceased. I treat these addresses as the "home addresses" of the fallen soldiers. To obtain addresses from the fatality records, I submit each epitaph to ChatGPT-4 with a prompt asking it to extract only the address from the text.[4] I employ a large-language model for this task due to the high variability in formatting of the epitaphs and addresses themselves, which renders rule-based and supervised machine learning methods inaccurate.[5] Once an address is obtained, I associate it with a longitude-latitude point via OpenStreetMap's geocoding service. I carry out the geocoding procedure on CWGC records pertaining to soldiers who fought

[4]In cases where addresses for a spouse and parents are both listed, I instruct ChatGPT to provide the spouse's address. An example address is listed in Appendix B.1.

[5]The prompt used to instruct ChatGPT to extract the addresses can be found in Appendix B.2.

for the United Kingdom and were native to British Isles. The CWGC database lists 839,626 records of deceased soldiers who fought for the United Kingdom, of these 504,065 (60%) both had a non-empty epitaph and were associated with a British Isles regiment. Of those, 446,295 (89%) were successfully geocoded.[6] Of the records that were successfully geocoded, 358,690 are located in England and Wales, representing between 60% and 70% of the total fatalities sustained by the two countries.[7]

A potential concern is the possibility that the sample of soldiers whose home addresses were successfully geocoded differs from the full population of deceased soldiers in some characteristic that is related to an outcome of interest. To test if this is the case, I compare the samples of geocoded and non-geocoded soldiers using observable characteristics present in CWGC. The results can be found in Appendix B.3. The sample of geocoded soldiers resembles the non-geocoded sample in terms of age at death, time of death, rank, and surname, though non-geocoded soldiers had more missing characteristics in the data than geocoded soldiers.

I use the geocoded fatality records to estimate the World War I fatality rate for each House of Commons parliamentary constituency in England and Wales as it was between 1918 and 1945. I first match each geocoded address to a House of Commons parliamentary constituency, using boundary files drawn from the Great Britain Historical GIS database (Southall and Aucott, 2009). Computing the fatality rate requires statistics on the number of soldiers from a constituency that participated in the war. Unfortunately, nearly two-thirds of UK WWI service records were destroyed in 1940, after an incendiary bomb was dropped on the War Office Record Store during the Blitz (The National Archives, 2026).

I therefore estimate the number of conscripted soldiers using the enlistment requirements in the Military Service Act of 1916 and census microdata from IPUMS (Ruggles et al., 2003). I first

[6]Failure to geocode a soldier with a non-empty epitaph happened for a variety of reasons. In some cases, family members had moved outside the United Kingdom. In others, the geocoder could not pin down addresses to locations more precise than a county.

[7]The precise number depends on estimates of Irish and Scottish losses in the war. In 1922, the War Office estimated that 702,000 soldiers from the British Isles perished during the conflict. Estimates of Irish war dead range from 30,000 to 40,000. Similar estimates place Scottish losses between 80,000 and 150,000. (Bunbury, 2014; UK War Office, 1922; Watt, 2019)

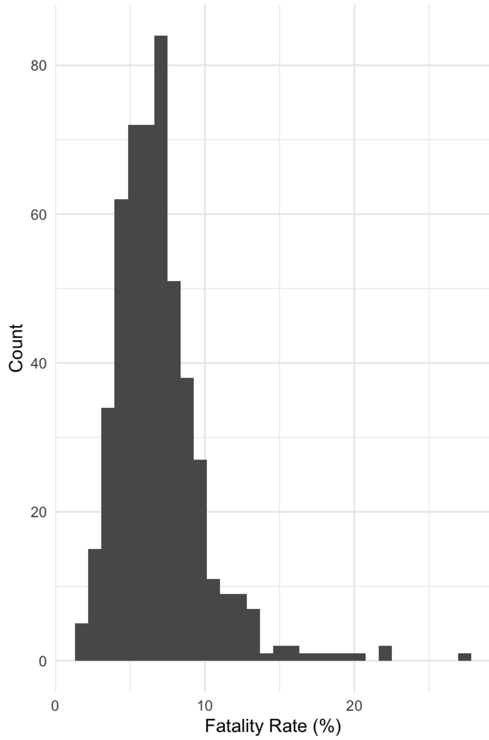


Figure 2: Fatality rate histogram

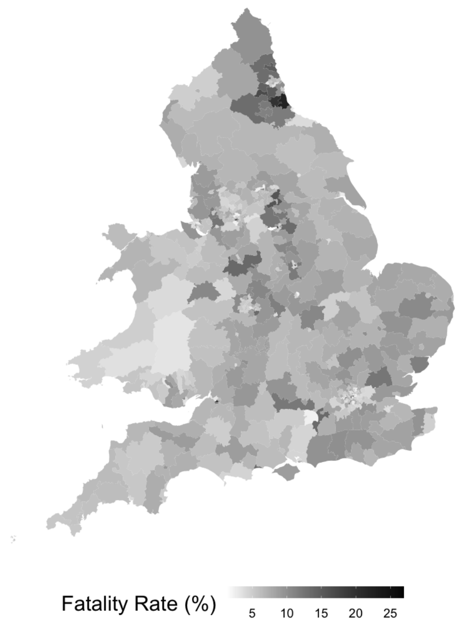


Figure 3: Fatality rate map

count the number of males between the ages of 11 and 46 in each civil parish in England and Wales, the smallest level of aggregation available in the 1911 census. This number represents the maximal set of men who could have been conscripted between 1916 and 1918. Using a 1918 "Schedule of Protected Occupations," I exclude all individuals employed in a reserved occupation that was deemed critical for the home front (Pattinson, 2016). This procedure estimates the number of males in each parish that were of draftable age and not employed in a reserved occupation, but does not account for health exemptions. The parish-level data is then aggregated to the level of the parliamentary constituency to compute the fatality rate. Figures 2 and 3 plot the histogram and geographical distribution of the conscript fatality rate. While the median constituency lost 6.3% of its draftable men, across all constituencies the rate ranged from 1.3% to 26.5%.

## 3.2 Outcomes

I study the effect of conscript fatalities on two outcome variables: the share of the vote received by left-wing candidates in general elections between 1918 and 1935, and roll-call votes on progressive taxation legislation in the House of Commons during the same period.

### 3.2.1 Left-wing vote share

To examine how WWI conscript fatalities influenced electoral support for left-wing political parties, I source constituency-level election results for the seven general elections between 1918 and 1935 from the database compiled by Eggers and Spirling (2014). To measure left-wing vote share, I sum the vote shares received by candidates that represented communist, socialist, or trade unionist parties.[8] Also included are independent candidates who self-identified as communist or socialist. Table 1 lists the average share of the vote received by left-wing candidates and the number of constituencies contested by at least one left-wing candidate in England and Wales.

| Election | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 |
|---|---|---|---|---|---|---|---|
| Mean left-wing vote share | 31.6 | 37.2 | 39.0 | 36.6 | 38.3 | 33.0 | 40.4 |
| Seats contested | 355 | 376 | 382 | 453 | 504 | 461 | 482 |

Table 1: Left-wing vote share and seats contested

### 3.2.2 Parliamentary votes on progressive taxation legislation

To measure legislator support for progressive taxation, I collected House of Commons voting records for 424 tax policy divisions (roll-call votes) that took place between 1918 and 1935, comprising 143,773 individual votes. The dataset was assembled via hand-collected records from *Hansard*, the official transcript of UK Parliamentary debate. The divisions relate to five categories of tax legislation, most of which are associated with the Finance Act, the UK's yearly headline budget legislation:

[8] A full list of parties categorized as left-wing is presented in Appendix B.4

1. **Capital levy**: Legislation on direct transfers of capital to the government. Includes proposals to impose a capital levy (a one-time wealth tax) and to nationalize industries.

2. **Customs duties**: Legislation on the implementation and level of tariffs on goods entering the United Kingdom. Excludes technical clauses clarifying the precise definition of the good on which the tariff is to be imposed, as well as legislation seeking to impose tariffs on goods from specific countries (e.g. Imperial Preference).

3. **Excise duties**: Legislation related to any form of non-tariff consumption tax. For instance, sales taxes on alcohol, tobacco and cinema tickets, as well the "betting duty," a tax on bookmaker profits.

4. **Income taxes**: Legislation related to rate and gradation of taxes on individual income. Also includes proposals that affect tax exemptions for individuals and their dependents.

5. **Wealth taxes**: Legislation pertaining to death and estate duties (forms of inheritance tax), the land value tax, and the Excess Profits Duty.

These five categories cover the universe of taxation legislation considered during the interwar period. Broad coverage of tax legislation is important because political parties carefully considered the ratio of tax revenues coming from direct and indirect taxation when constructing their tax platforms (Daunton, 2002). For instance, the Labour Party favored a progressive tax platform that emphasized direct taxation. It consequently stridently opposed consumption taxes and customs duties, seeing them as regressive instruments that raised the cost of living for workers. In contrast, the Conservative Party preferred to finance the government through tariffs and other sources of indirect taxation.

The salience of different forms of taxation varied over time. Figure 4 plots the proportion of roll-call votes by topic in each parliament. While roughly a fourth of tax divisions were concerned with the income tax, there was significant variation in the discussion of other forms of taxation. Divisions on wealth taxes spiked in the Parliaments of 1918-1922, 1923-1924, and 1929-1931, a

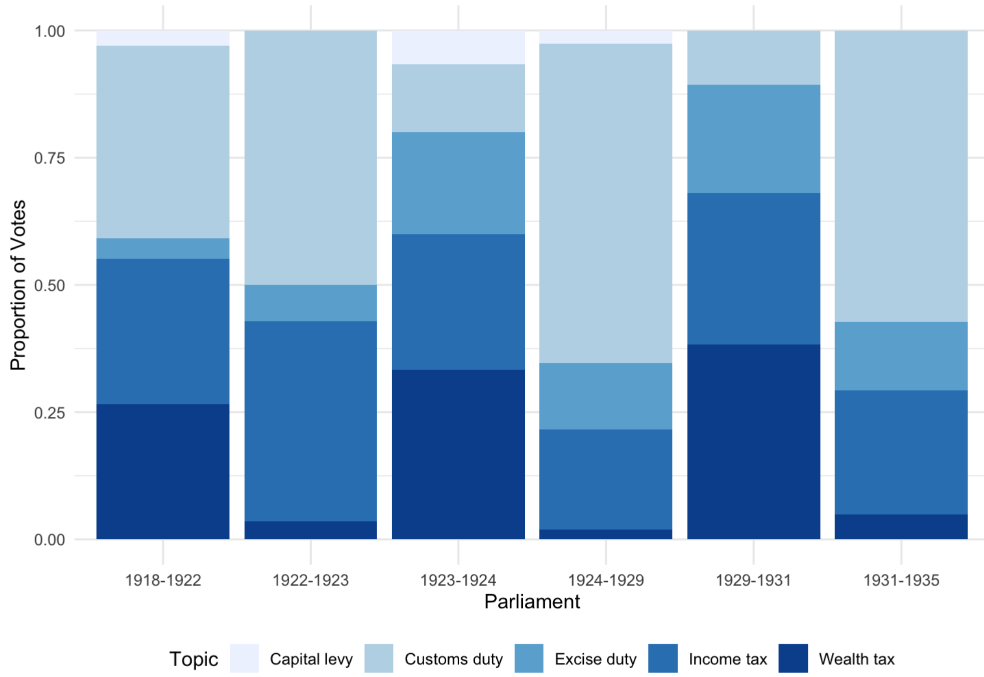


Figure 4: Divisions by topic and parliament

consequence of debates over the abolition of Excess Profits Duty and the (re-)introduction of the land value tax. Divisions on customs duties also became more frequent after the start of the Great Depression in 1929, comprising more than half of all taxation votes in 1931 to 1935.

For each House division, I code whether an individual "Aye" or "No" vote is considered to be in support of progressive taxation, and then record each MP's position. An individual vote is categorized as "pro-progressive taxation" if the consequence of the corresponding legislation's passage would result in greater tax progressiveness. For instance, an "Aye" vote may be pro-progressive if it is associated with legislation that increases the highest marginal tax rate. An example of a pro-progressive "No" vote is one against legislation that seeks to abolish inheritance taxes. Full coding rules and example legislation can be found in Appendices B.5 and B.6.

## 3.3 Plausible exogeneity of conscript fatalities

Interpreting a correlation between conscript fatalities and support for progressive taxation as causal requires addressing the existence of potential confounding variables. One such variable is socioeconomic status. In the early 20th century, Labour and other trade unionist parties drew much of their support from industrial and mining regions. Conscripts from industrial areas were often less healthy than conscripts from rural areas, due to a combination of harsh labor conditions, poor sanitation, and exposure to pollution (Davenport, 2020). Worse overall health may have made conscripts hailing from industrial regions less likely to survive the exacting physical demands of war (Kriner and Shen, 2010). As a result, it is possible that greater electoral support for left-wing candidates in locales with higher fatality rates stems from preexisting socioeconomic conditions.

Another channel through which socioeconomic status might influence both conscript fatality rate and left-wing vote share is through decisions about force employment. If officers intentionally used conscripts from lower socioeconomic backgrounds as "cannon fodder" by placing them in more dangerous combat situations, we would again observe a higher fatality rate in constituencies that had a preexisting inclination to support left-wing candidates (Huff, Min and Schub, 2024; Mueller, 1991). Moreover, although near-universal male conscription made it unlikely that ideological considerations led soldiers to select into military service, it could still be the case that soldiers' principles influenced their likelihood of dying in combat. For instance, patriotic soldiers may have volunteered for hazardous assignments at greater rates. A constituency's prewar partisan lean could affect both postwar political behavior and the propensity to take fatal risks during war (Rozenas, Talibova and Zhukov, 2024).

To mitigate these potential issues, my empirical design exploits the quasi-random assignment of conscripts to battlefields arising from the organizational structure of the British Army. Infantry recruitment in the UK during WWI was primarily conducted through line regiments with geographical affiliations. Line regiments were allotted administrative counties to draw soldiers from, and consequently followed county boundaries, though in some cases a recruiting ground contained multiple sparsely populated counties, as visualized in Figure 5 (Imperial War Musuem, 2025).[9] Regiments were composed of battalions, groupings of roughly 1,000 men that were the pri-

[9]Some specialized regiments, such as the Royal Engineers, recruited over the whole of the United Kingdom.

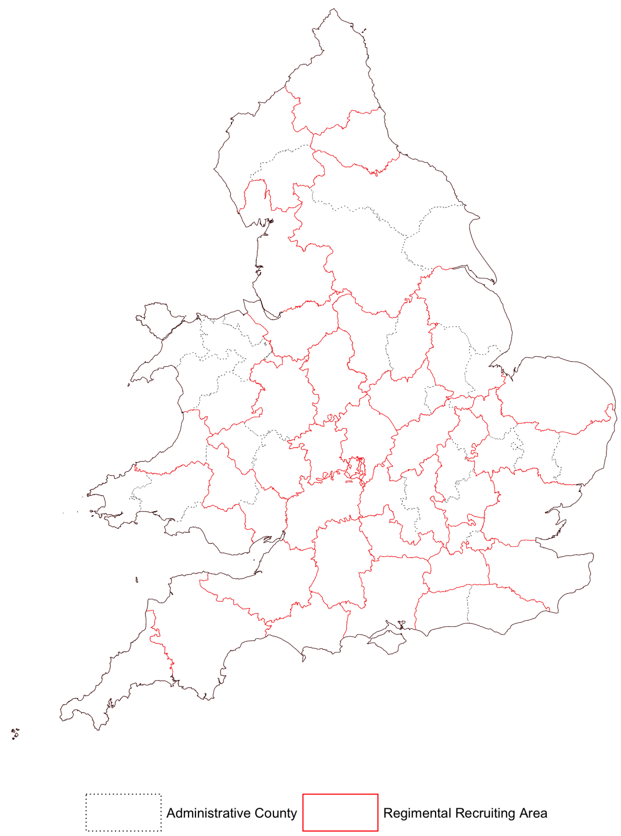


Figure 5: WW1 regimental recruiting areas and 1911 administrative counties

mary tactical units in the British Army, with soldiers in a battalion all deployed to the same front (James, 2012). Although the Army permitted "Pals Battalions," or locally-organized battalions, in the initial stages of the war, with the introduction of conscription all recruits were pooled together and assigned to a battalion randomly or by greatest need (French, 2005). Therefore, which battle a new conscript was sent to was determined by a combination of when a conscript turned 18 years old (becoming eligible to be drafted) and the distribution of solider casualties across battles in the weeks preceding the new conscript's completion of basic training. Earlier combat lethality itself is also likely driven primarily by near-random events on the battlefield. Given this organizational structure, it is unlikely that systematic prewar differences affected the fatality rate of local conscripts within the same regiment.

To assess the validity of these claims, I take a similar approach to De Juan et al. (2024) and conduct a series of balance tests that demonstrate WWI conscript fatalities are uncorrelated–

conditional on administrative county–with a range of demographic, economic, and political features of parliamentary constituencies. To do so, I estimate bivariate regressions of the WWI conscript fatality rate on several constituency-level variables with administrative country fixed effects. Using fixed effects at the level of the administrative county rather than line regiment captures more granular unobserved factors than recruiting grounds, helping account for the effect of Pals Battalions earlier in the war. Figure 6 plots the obtained coefficients and 95% confidence intervals computed using robust standard errors clustered at the administrative county level.

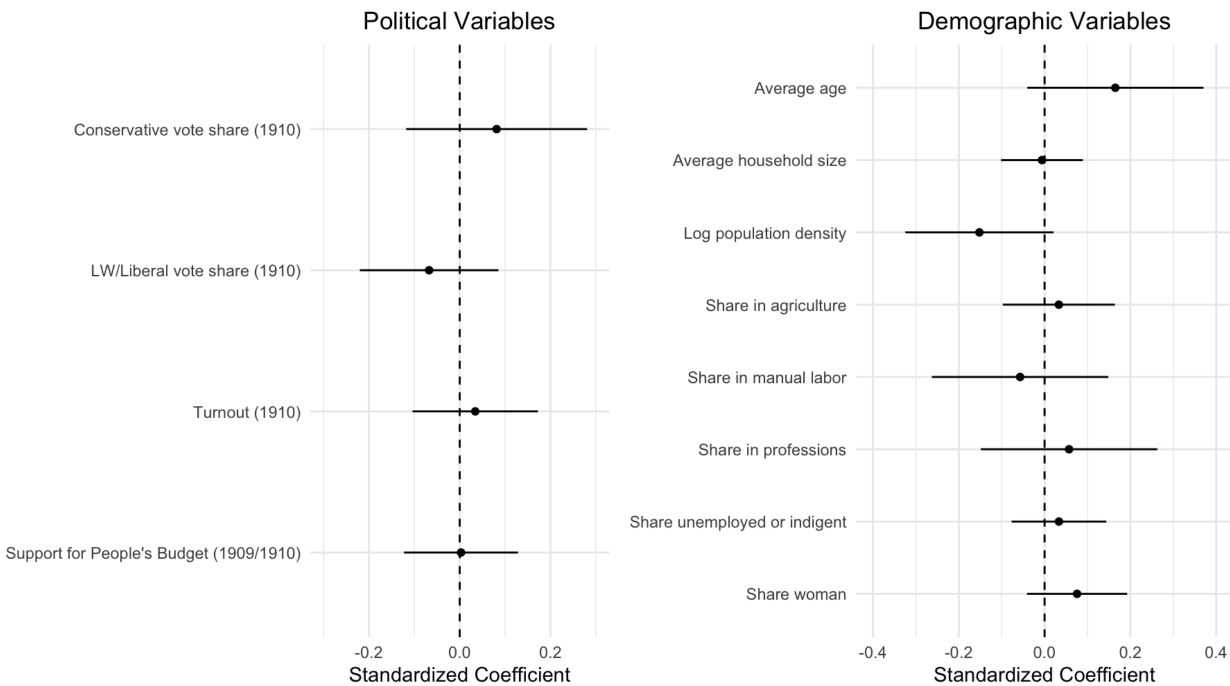


Figure 6: Placebo tests for conscript fatalities on prewar outcomes

The left panel of Figure 6 shows that the WWI conscript fatality rate does not have a statistically significant correlation with Conservative Party vote share, left-wing or Liberal Party vote share, turnout in the 1910 UK general election, and MP votes in favor of the 1909/1910 "People's Budget."[10] The People's Budget, the colloquial name for the Finance Act of 1909/1910, was the first UK budget to include redistribution as goal, introducing the country's first progressive income tax

[10] I sum the vote shares of left-wing and Liberal candidates to account for the fact that left-wing parties contested only 62 of the 670 seats in the House of Commons during the 1910 election (Eggers and Spirling, 2014). Moreover, between 1885 and 1910, the Liberal Party had a policy of sponsoring trade unionist candidates, meaning about a third of left-wing candidates were also affiliated with the Liberal Party (Powell, 1986).

and a land tax. The lack of correlation suggests that prewar political behavior is unlikely to confound the effect of WW1 fatalities on support for progressive taxation. Similarly, the right panel of Figure 6 reveals that the WWI conscript fatality rate is not correlated with constituency-level economic and demographic characteristics drawn from 1911 UK census.[11] These include variables capturing the demographic structure of a constituency, such as the share of women, average age, average household size, and population density, as well as those characterizing economic activity, such as occupational mix (i.e., share of workers employed in agriculture, the professions, and industrial labor). These results offer evidence that WWI conscript fatalities are unrelated to prewar support for progressive taxation and other potential sources of political preferences.

# 4 Empirical analysis

## 4.1 Electoral outcomes - empirical strategy

To assess the effect of World War I conscript fatalities on left-wing vote share during the interwar period, I estimate the following equation via ordinary least squares for each general election:

$$leftShare_{ic} = \alpha fatalityRate_i + \beta candidates_i + \gamma^T \mathbf{X}_i + \delta_c + \epsilon_{ic} \tag{1}$$

where $leftShare_{ic}$ is the percentage of the vote received by a left-wing candidate in parliamentary constituency $i$ within administrative county $c$, $fatalityRate_i$ is the WWI fatality rate of conscripted soldiers in constituency $i$, and $\epsilon_{ic}$ is an error term. The variable $candidates_i$ represents the number of candidates contesting constituency $i$. The term $\delta_c$ is an administrative county fixed effect. Finally, $\mathbf{X}_i$ is a vector of controls, consisting of prewar demographic, geographic, and political characteristics of constituency $i$. The coefficient of interest is $\alpha$, which represents the expected change in the share of the vote received by left-wing parties resulting from a one-percentage point increase in the fatality rate of WWI conscripts.

[11]Descriptive statistics for all constituency-level data can be found in Appendix B.7.

I include the number of candidates contesting a race as a control variable to ensure vote shares are comparable across races. During the period under study, some races featured as many as seven candidates, with individual vote shares generally decreasing with the number of candidates running for office. The number of candidates competing for office is a post-treatment variable, which can introduce collider bias if part of the effect of war fatalities on left-wing vote share is mediated through the number of candidates (Rosenbaum, 1984). However, there are reasons to believe that conscript fatality rates did not systematically influence the number of candidates running for office. First, for much of the interwar period, the three major parties stood candidates for election in nearly all constituencies. Variation in the number of candidates seems to instead be driven primarily by party fractures or idiosyncratic regional political movements (Eggers and Spirling, 2014). Second, in Appendix C.3, I empirically test whether fatality rates affected number of candidates choosing to run for office and find no evidence of an association. Consequently, I treat the number of candidates as resembling a pre-treatment variable that does not bias the coefficient on the conscript fatality rate (Pepinsky, Goodman and Ziller, 2024). Out of caution, in the Online Appendix, I also estimate models excluding control variables and using electoral margin as an alternative outcome.

## 4.2 Effect of war fatalities on electoral outcomes

Table 2 displays the coefficients obtained from estimating regressions of left-wing vote share on the WWI conscript fatality rate. Columns 1 through 7 report the coefficients on the conscript fatality rate obtained from estimating Equation 1 separately for each general election between 1918 and 1935. Column 8 reports the coefficient obtained from estimating Equation 1 on data that pools together all elections in the interwar period. The reported specifications include administrative county fixed-effects and a suite of prewar constituency-level demographic and political variables. Standard errors are clustered by administrative country.

The WWI conscript fatality rate is positively associated with the share of the vote received by candidates from economically left-wing parties in all specifications. This association is statisti-

| | DV: Left-wing vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | 0.36** | 0.34** | 0.66*** | 0.32 | 0.43** | 0.15 | 0.33** | 0.37*** |
| | (0.16) | (0.16) | (0.23) | (0.24) | (0.18) | (0.13) | (0.16) | (0.13) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. left-wing vote share (%) | 31.92 | 38.30 | 40.19 | 37.35 | 38.69 | 33.09 | 40.83 | 37.32 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 47 | 54 | 51 | 54 | 57 | 53 | 56 | 58 |
| N | 327 | 354 | 363 | 422 | 481 | 434 | 451 | 2832 |
| $R^2$ | 0.46 | 0.69 | 0.69 | 0.74 | 0.78 | 0.75 | 0.76 | 0.59 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table 2: Effect of WW1 fatalities on left-wing vote share

cally significant at the 5% level for the pooled specification and all individual elections except for 1924 and 1931. During the interwar period as a whole (Column 8), a one-percentage-point rise in the conscript fatality rate increased the share of the vote received by left-wing candidates by 0.37 percentage points. This result is illustrated in Figure 7, which depicts the average left-wing party vote share and WWI fatality rate for all constituencies in England and Wales. The maps show that left-wing political parties consistently obtained greater shares of the vote in locations with higher WWI fatality rates, particularly in the north of England and southern Wales.

The individual election models (Columns 1 through 7) show that the effect of the fatality rate on left-wing electoral support persisted until 1935 but with in notable variation in magnitude by election. For instance, the coefficient is largest for elections that took place in the five years after the end of the war. The magnitude of the effect also increases during the first three elections, peaking in 1923, where a one percentage point increase in the war fatality rate is associated with a 0.66 percentage point increase in left-wing party vote share. This notably coincides with the Labour Party's most successful election to that point, where it obtained 191 seats and formed its first government. Afterwards, the magnitude of the effect of war fatalities declines, but remains statistically significant in 1929 and 1935.

The effect of war fatalities may appear small, especially considering that during the period

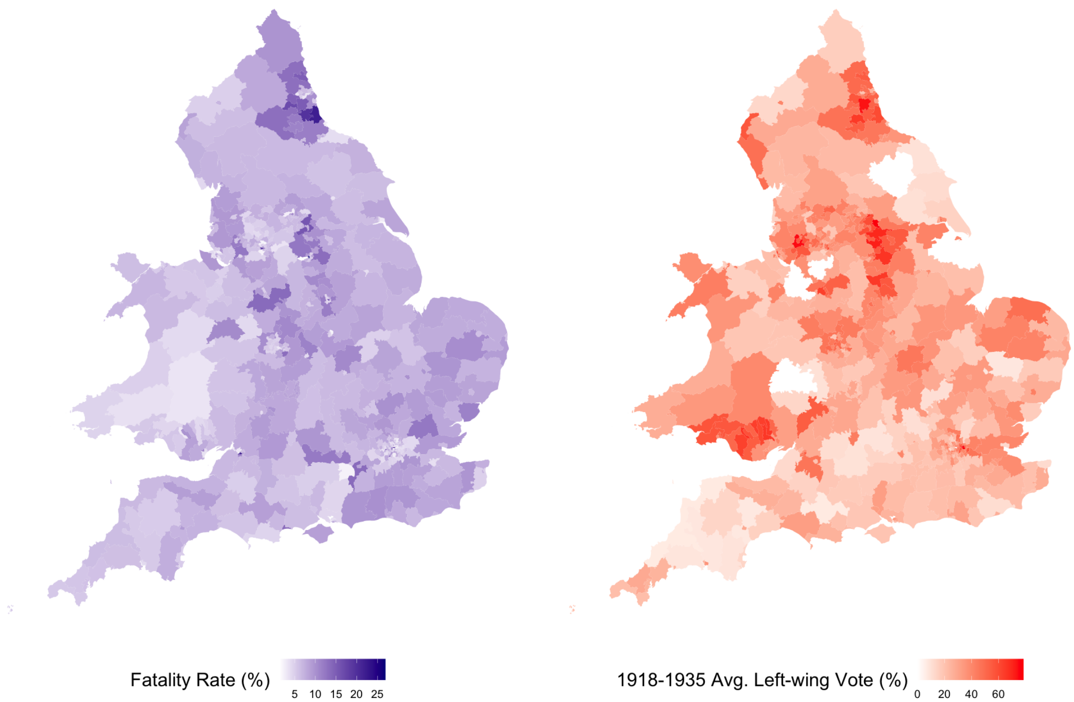


Figure 7: Constituencies by left-wing vote share and WWI fatality rates

under study the average margin of victory across all races was 21%. However, these averages are skewed by large margins in safe seats, obscuring the fact that roughly one-sixth of races during the interwar period were decided by a margin of less than 5%. In these races even small changes in vote share could change the outcome. To evaluate the implications of war fatalities for control of the House of Commons, I use the estimates obtained in Table 2 to compute counterfactual election outcomes for all constituencies in England and Wales.[12] The results are plotted in Figure 8. The counterfactual estimates reveal that in 1923, incrementing the war fatality rate by one standard deviation (3%) would have flipped 22 seats to left-wing parties, while a two-standard-deviation increase (6%) would have resulted in an additional 53 left-wing candidates winning office. While the counterfactuals for other elections do not produce as dramatic swings in seats, they still suggest that changes in vote share resulting from war fatalities could have significant consequences for the composition of the House of Commons. For example, a one-standard deviation increase

[12]The counterfactuals assume that the vote share gained by left-wing parties is drawn equally from the shares of the other parties.

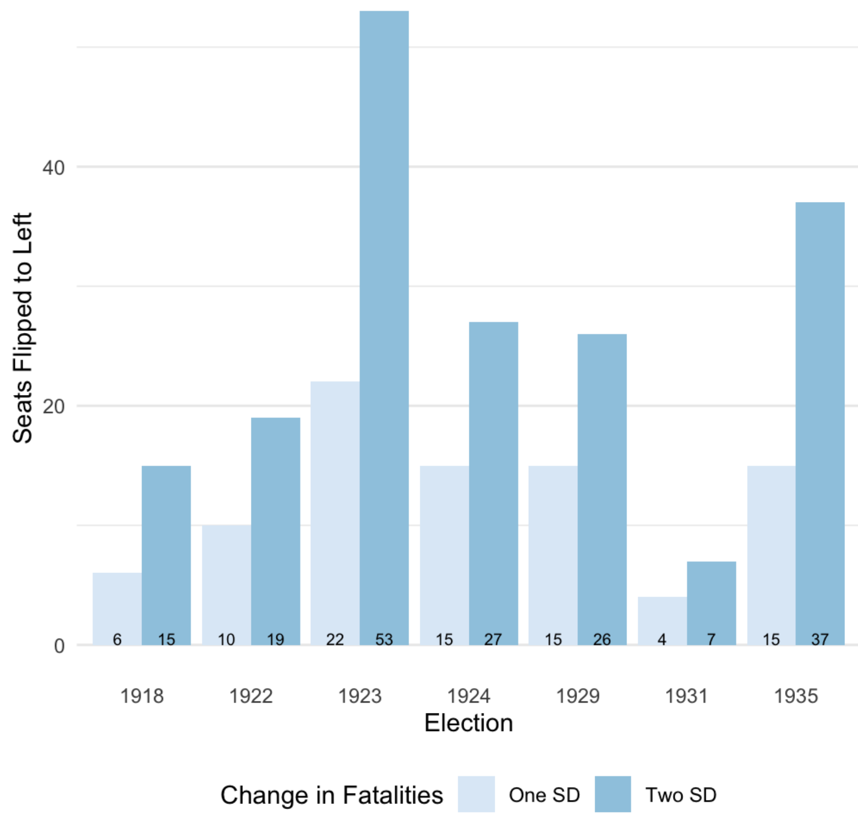


Figure 8: Counterfactual parliamentary seat changes

in war fatalities would have put Labour six seats shy of an outright majority in 1929, even disregarding potential seat changes in Scotland and Northern Ireland, a feat that the party would not truly achieve until 1945.

The results of the regression analysis indicate that voters in constituencies with greater WW1 conscript fatalities were more likely to support economically left-wing candidates, consistent with the compensatory theory. In the Online Appendix, I offer additional evidence that suggests support for left-wing political parties in high fatality constituencies was driven by the increased salience of fairness considerations. First, I examine how war fatalities affected the vote shares received by candidates representing the Conservative and Liberal Parties. I find that conscript fatalities had no effect on support for the Conservative Party but are associated with reduced support for the Liberal Party in some election years. As the Conservative Party drew support from business owners, i.e., those that may have profited from the war and would be hurt if progressive taxation schemes were implemented, it is consistent with the compensatory theory that local fatalities did not draw voters away from the Conservatives. In contrast, some erstwhile supporters of the Liberals became disillusioned with the party's moderate stance after the war and switched their vote to more left-wing candidates, contributing to the decline of the Liberal Party during the 1920s (Adelman, 2014). Second, while war fatalities do not appear to have impacted Conservative vote shares, they did make Conservative candidates electorally vulnerable. I find that a one-percentage-point increase in the WW1 fatality rate increased left-wing candidates' electoral margins by 0.68 percentage points on average. Former Liberal voters switching to Labour in areas with high fatalities meant that Conservative candidates received less advantage from vote splitting by Labour and Liberal candidates.

Additionally, I repeat the vote share analysis while conditioning on 1910 left-wing vote share. As socialist and trade unionist movements were still nascent in 1910, introducing this control variables limits the analysis to constituencies that left-wing parties regarded as being worth contesting. These constituencies were largely manufacturing and mining hubs with disproportionate exposure to unfair wartime government policies, such as the outlawing of striking. I find that war fatalities had a larger effect on left-wing vote share in these constituencies after the war, suggesting that the "conscription of wealth" held greater appeal in locales where a sense of unfairness was more keenly felt. Finally, I show that overall results are robust to alternative model specifications.

### 4.3 Parliamentary voting - empirical strategy

Having shown that conscript fatalities are associated with electoral support for left-wing political parties, I now examine whether conscript fatalities impacted the passage of progressive taxation legislation in the House of Commons. Specifically, I test whether members of the House of Commons representing constituencies with higher World War I fatality rates were more likely to vote in favor of progressive taxation legislation. To do so, I estimate the following model via ordinary least squares

$$proTaxVotes_{ictp} = \alpha fatalityRate_i + \gamma^T \mathbf{X}_i + \delta_c + \nu_t + \lambda_p + \epsilon_{ictp} \tag{2}$$

where $i$ indexes constituencies, $c$ indexes administrative counties, $t$ indexes years, and $p$ indexes political parties. The variable $proTaxVotes_{ictp}$ is the number of times the MP of party $p$ representing constituency $i$ in county $c$ during year $t$ voted in favor of progressive taxation legislation. Similar to the vote share specifications, $fatalityRate_i$ is the WWI fatality rate in constituency $i$, $\mathbf{X}_i$ is a vector of prewar controls, $\delta_c$ is an administrative county fixed effect, and $\nu_t$ is a year fixed effect. New to this specification is a party fixed effect, $\lambda_p$. The party fixed effect means that the models examine within-party variation in voting behavior to ensure that any observed relationship is not being driven by more left-wing candidates winning office.

## 4.4 Parliamentary voting - results

The coefficients obtained from estimating Equation 2 are displayed in Table 3. Columns 1 through 6 report coefficients obtained from fitting the model to each Parliament between 1918 and 1935 separately, while Column 7 reports the coefficient for the interwar period as a whole. Reported standard errors are clustered by administrative county. The results show that there is an overall positive association between the WWI fatality rate of a constituency's conscripts and the frequency with which the MP representing that constituency voted in favor of progressive taxation. For the entire interwar period (Column 7), a one-percentage point increase in the local fatality rate is associated with an additional .07 votes cast in support of progressive taxation per MP per year. This finding indicates that conscript fatalities facilitated the introduction of progressive taxation schemes in interwar Britain.

Table 3 also provides evidence that electoral pressure was the mechanism through which conscript fatalities affected legislator behavior. As was the case for left-wing vote share, the effect of fatalities on MP support for progressive taxation varies over time. Notably, the effect is

| | DV: Num. pro-progressive tax votes | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1918-1922 | 1922-1923 | 1923-1924 | 1924-1929 | 1929-1931 | 1931-1935 | Pooled |
| Fatality rate (%) | $0.10^{**}$ | $0.12^{*}$ | 0.05 | 0.09 | $0.14^{**}$ | $-0.01$ | $0.07^{***}$ |
| | (0.05) | (0.06) | (0.03) | (0.07) | (0.06) | (0.02) | (0.02) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Party FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| DV Mean: | 3.47 | 7.31 | 5.24 | 7.58 | 6.36 | 1.92 | 4.69 |
| Num. clusters: | 57 | 58 | 58 | 58 | 58 | 58 | 58 |
| N | 1887 | 493 | 499 | 1974 | 1460 | 2344 | 8866 |
| $R^2$ | 0.68 | 0.84 | 0.61 | 0.74 | 0.71 | 0.68 | 0.63 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table 3: Effect of WW1 fatalities on MP support for progressive taxation

largest in parliaments following elections where war fatalities provided the greatest benefit for left-wing candidates: 1918, 1922, and 1929. In these three parliaments, a one percentage point rise in the war fatality rate increased the frequency of MP support for progressive taxation by around twice as much as for the interwar period as whole. The fact that MPs representing high fatality constituencies were more likely to vote in favor of progressive taxation legislation when fatalities had the greatest electoral salience implies that legislators were responding to constituent preferences. Importantly, the effect of conscript fatalities is detectable even with the inclusion of party and year fixed effects, suggesting that the increased support for progressive taxation was not simply due to more left-wing politicians taking office or national trends in politics.

Evidence of a within-party effect of war fatalities on MP behavior is notable given the high degree of party cohesion during the early 20th century–past work has documented that UK political parties at this time exhibited almost no within-party factional conflict (Eggers and Spirling, 2014; Cox, 2005). Strictly enforced party discipline during roll-call votes likely tempered MPs responses to popular demands arising from war fatalities. For instance, party whips persuaded, bribed, threatened, and even blackmailed MPs to follow the party line (Troughton, 2023). MPs that disregarded voting instructions were subject to various punishments, which ranged from being passed over for ministerial posts to expulsion from the party (Pollock, 1930). As whipping generally succeeded in enforcing party cohesion, it seems likely that much of the effect found in the prior analyses is driven by defectors from the party position.

To get a better look at how local war fatalities influenced MP voting behavior within political parties, I estimate Equation 2 separately for the three major parties. The results are shown in Table 4. I find that war fatalities had a significant effect on Conservative MPs propensity to vote in favor of progressive taxation. While the coefficient of 0.03 is smaller than the overall effect found in Table 3, the size of the effect is notable in light of the fact that Conservative MPs only voted in support of progressive taxation an average of 0.67 times per year. To illustrate the relative effect size, a one standard deviation rise in the conscript fatality rate would increase the frequency of pro-progressive tax votes cast by the average Conservatives by 13.4%. In contrast, I find no evidence that Labour and Liberal MPs representing high-fatality districts were more likely to support progressive taxation legislation.

| | DV: Num. pro-progressive tax votes | | |
|---|---|---|---|
| | Conservative | Labour | Liberal |
| Fatality rate (%) | 0.03*** | 0.10 | 0.02 |
| | (0.01) | (0.07) | (0.29) |
| Controls | ✓ | ✓ | ✓ |
| County × Year FEs | ✓ | ✓ | ✓ |
| DV Mean: | 0.67 | 15.07 | 8.68 |
| No. clusters: | 52 | 37 | 51 |
| N | 4492 | 1918 | 649 |
| $R^2$ | 0.64 | 0.80 | 0.84 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table 4: Effect of WW1 fatalities on MP support for progressive taxation by party

The party-level results provide additional evidence that electoral pressure from conscript fatalities induced MPs to adopt stances supporting progressive taxation. Labour and radical Liberal MPs supported progressive taxation for ideological reasons and benefitted from pro-redistribution shifts in constituent preferences caused by war fatalities. There would be little incentive for Labour and Liberal MPs to change stance on progressive taxation issues, resulting in less variation in voting behavior by fatality rate. Conservatives faced a different strategic calculus. As shown in Section 4.2, Conservatives in high-fatality constituencies became electorally vulnerable to challengers from the left. Conservative MPs that wished to retain their seats may have adopted pro-progressive taxation positions, even if this meant bucking the party line on votes. A closer examination of the constituency characteristics of Conservative MPs who defected from the party line lends credence to this explanation. Conservative MPs that defected at least once from the party line to vote in favor of progressive taxation legislation were 10% more likely to represent a constituency with an above-median WWI fatality rate than their colleagues that never defected. In addition, defection appears to have been more frequent among MPs representing constituencies with high fatality rates: defectors were 25% more likely to represent constituencies in the top decile of war fatalities.

| | DV: Num. pro-progressive tax votes | | | | |
|---|---|---|---|---|---|
| | Income tax | Wealth tax | Capital levy | Excise duty | Customs duty |
| Fatality Rate (%) | 0.01 | 0.04*** | 0.00 | 0.01 | 0.02* |
| | (0.01) | (0.01) | (0.00) | (0.01) | (0.01) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ |
| County × Year FEs | ✓ | ✓ | ✓ | ✓ | ✓ |
| Party FEs | ✓ | ✓ | ✓ | ✓ | ✓ |
| DV Mean: | 1.40 | 1.94 | 0.33 | 0.98 | 2.13 |
| Num. clusters: | 58 | 58 | 58 | 58 | 58 |
| N | 7040 | 4150 | 2064 | 5565 | 8253 |
| $R^2$ | 0.70 | 0.70 | 0.85 | 0.57 | 0.64 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table 5: Effect of WW1 fatalities on MP support for progressive taxation by topic

Finally, in Table 5 I analyze how WW1 conscript fatalities influenced MP support for particular types of progressive taxation legislation. I find that fatalities had the largest effect on support for wealth taxes. This suggests that MPs sought to assuage constituents' fairness concerns by directly targeting the accumulation of wealth from the war, which policies like the Excess Profits Duty were explicitly designed to do (Billings and Oats, 2014). Taken together, the results presented in this section offer evidence that local conscript fatalities pushed MPs to support progressive taxation, particularly among Conservative MPs and for wealth taxes.

# 5 Conclusion

This research note finds evidence that local World War I conscript fatalities increased support for progressive taxation in interwar Britain. Constituencies with higher conscript fatality rates provided greater electoral support to candidates representing left-wing political parties. These effects were strongest in elections immediately following the war and in constituencies where left-wing parties had an established presence prior to the war. In addition, members of Parliament representing districts that experienced more fatalities voted in favor progressive taxation legislation more frequently. This effect was pronounced among members of the Conservative Party and on legislation relating to wealth taxes, consistent with anecdotal evidence that Conservatives felt electoral pressure to accommodate voter demands for a conscription of wealth.

The findings of this note offer three empirical contributions to understanding of the mechanisms of the compensatory theory. First, using a two-part empirical analysis, this study outlines an electoral mechanism through which individual fairness preferences affected the introduction of compensatory tax policy in the UK. Specifically, MP support for progressive taxation was induced by shifts in voter support toward economically left-wing parties. Second, this note illustrates the persistent effect of fairness concerns on postwar tax policy. The legacy of wartime conscription continued to be salient during elections and taxation debates into the 1930s. Finally, this note uses a new research design employing cross-sectional variation in exposure to unfair government policy to confirm and strengthen confidence in the predictions of the compensatory theory.

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# Online Appendix

## Table of Contents

# A A model of debt, democracy, and war fatalities

In this section, I present a simple formal model to study the effect of debt and war fatalities on tax progressiveness, based on existing models of redistributive politics (Cox and McCubbins, 1986; Dixit and Londregan, 1996; Lindbeck and Weibull, 1987; Roemer, 2009). Two office-motivated political parties compete in a probabilistic voting setting by proposing taxation platforms that distribute the burden of paying off government debt between social classes. The effect of war fatalities enters the model in two ways: increased class consciousness among the poor and ideological bias toward a party representing the working class, capturing the claim of Scheve and Stasavage (2016) that government policies that disproportionately impact the poor, such as conscription, generate support for taxing the rich as compensation.

The model illustrates how the debt incurred from war can itself drive progressiveness, thus potentially confounding a proposed relationship between mass mobilization for war and progressive taxation. This arises because political parties face a "vote-revenue" trade-off: taxing a social class helps pay off debt but drives away voters (Cox and McCubbins, 1986). Under conditions of high economic inequality, taxing the wealthy yields higher revenues at a lower electoral cost. The model also demonstrates that ideological bias in favor of a party representing the poor also induces candidates to propose more progressive tax platforms. This suggests an empirical approach to test compensatory arguments for tax progressiveness that avoids the confounding influence of fiscal necessity. While war debt is national issue, impacting all constituencies in common, constituencies vary in the degree to which their conscripts died during the fighting. If war fatalities generate demands for compensation and therefore ideological support for left-wing parties, we can leverage local variation in war deaths to verify the compensatory theory of tax fairness.

## A.1 Setup

Consider a society inhabited by a continuum of individuals, where the population is normalized to 1. Individuals are stratified into two socioeconomic classes, wealthy and poor, where poor individuals constitute a strict majority of size $q > \frac{1}{2}$. Wealthy individuals earn an income of $y_w$, while poor individuals earn $y_p$, with $y_w > y_p$.

Income is taxed at a group-specific tax rate $t_g$ where $g \in \{p, w\}$ and $t_g \in [0, 1]$. I assume $t_g$ fully determines the tax burden of group $g$ (i.e., includes both direct and indirect taxation) and is net of transfers. I further assume that income thresholds for group-specific taxes are fixed. These group-specific tax rates capture in a simplified way the fact that tax policies can be designed to have different impacts on socioeconomic groups. For instance, income tax regimes can be made progressive to account for the wealthy's greater ability to pay, while consumption taxes can be regressive because the poor consume more as a proportion of income. Individual preferences are given by post-tax income $(1 - t_g)y_g$. The utility of an individual from group $g$ is given by the function $u_g(t_g)$, where $u_g$ is decreasing in its argument and concave. This formulation imposes the assumption that individuals of both classes always prefer a lower tax on their own income, but are indifferent to the tax rate levied on members of the other social class.

Political competition takes place between candidates of two political parties, denoted $L$ and $R$, who compete for office by proposing tax platforms $\mathbf{t}^L = (t_p^L, t_w^L)$ and $\mathbf{t}^R = (t_p^R, t_w^R)$. As is common in probabilistic voting models, I assume that the candidates commit to implement their proposed platform should they be elected. The candidates are office-motivated and obtain no benefit from setting policy beyond its electoral implications. However, for purposes of exposition I assume that the parties also profess an ideology, with party $L$ claiming to represent poor individuals and party $R$ claiming to represent wealthy ones. The candidates' proposed tax platforms are constrained by the existence of government war debt of size $D > 0$. Importantly, candidates' tax platforms must be feasible, in that they raise sufficient funds to meet debt obligations.

**Definition 1 (Feasible tax platform)** *A tax platform $(t_p, t_w)$ is feasible if it satisfies*

$$t_p q y_p + t_w(1-q) y_w \geq D \tag{3}$$

I assume that the debt is small enough that there exist feasible tax platforms, but large enough that it cannot be paid off solely by taxing a single group. This ensures that feasible tax rates must take strictly positive values. I also assume that any tax revenue in surplus of debt obligations is rebated back to the population, so constraint 3 binds with equality.[13]

A candidate is elected to office via simple majority rule, where each individual has a single vote. Individuals decide which candidate to vote for based on how each candidate's taxation platform affects their consumption and idiosyncratic ideological preferences regarding the candidates' positions. Specifically, individual $i$ in group $g$ will vote for the candidate of party $L$ if

$$u_g(t_g^L) > u(t_g^R) + \sigma^{ig} + \delta(\alpha)$$

where $\sigma^{ig}$ is an individual-level parameter that captures individual $i$'s bias toward candidate $L$. I assume that $\sigma^{ig}$ is distributed uniformly in group $g$ over $\left[-\frac{1}{2\phi_g}, \frac{1}{2\phi_g}\right]$. The term $\delta(a)$ represents an electorate-wide preference shock against party $R$, and is distributed uniformly over $\left[-\alpha - \frac{1}{2\psi}, -\alpha + \frac{1}{2\psi}\right]$ where $\alpha > 0$ and $\psi > 0$. The candidates do not know the exact value of $\delta(a)$ but are aware of its distribution. I assume that $\phi_g$, $\alpha$, and $\psi$ take values that ensure the distributions are wide enough to avoid boundary issues in equilibrium. This requires the following assumption.

**Assumption 1** *To avoid boundary issues in equilibrium, I assume that $\phi_g < \frac{1}{2}\left(\frac{\psi}{\psi+1}\right)$ for $g \in \{p, w\}$ and $\alpha < \frac{1}{2\psi}$.*

The variable $\alpha$ is the average bias against $R$ and is meant to represent the political effect of war fatalities or any other intervention that may influence ideological support for the left. I also assume that $\phi_p$, the degree of ideological homogeneity among the poor, is an increasing function of $\alpha$. These assumptions capture two ways in which mass mobilization for war influences conceptions of just taxation. First, there is evidence that the common experience of conscription among the working classes and perception of unfair treatment increased class consciousness, making voters more ideological homogeneous and sensitive to taxation (Daunton, 2002). Such a class-wide shift in preferences can be modeled as an increase in $\phi_p$. Similarly, the inclusion of $\delta(\alpha)$ is intended to model the contention of Scheve and Stasavage (2016) that government policies which disproportionately harm the poor, such as conscription, induce concerns about fairness in future policy that are distinct from preferences about future consumption. A desire for government policy to redress this past inequity should consequently manifest as a shift in

[13] As a result, electoral competition will take place over one policy dimension, with $t_w$ as a residual.

preferences toward party $L$, the party that claims to represent the poor. In short, war fatalities influence voting in two ways: first, by creating a general preference shift toward the party of the poor across all voters, and second, by making poor voters a more cohesive bloc.

Under these assumptions and given party platforms $\mathbf{t}^L$ and $\mathbf{t}^R$, the share of individuals from group $g$ that vote for party $L$ is

$$s_g^L(\mathbf{t}^L, \mathbf{t}^R) = \frac{1}{2} + \phi_g(u_g(t_g^L) - u_g(t_g^R) - \delta(\alpha))$$

It follows that $L$'s total share of the vote is

$$s^L(\mathbf{t}^L, \mathbf{t}^R) = \frac{1}{2} + q\phi_p(u_p(t_p^L) - u_p(t_p^R) - \delta(\alpha)) + (1-q)\phi_w(u_w(t_w^L) - u_w(t_w^R) - \delta(\alpha)) \tag{4}$$

Finally, party $L$ wins the election if it obtains a majority of the vote. Given the uncertainty about the true value of $\delta(\alpha)$, this happens with probability

$$P\left(s^L(\mathbf{t}^L, \mathbf{t}^R) > \frac{1}{2}\right) = \frac{1}{2} + \psi\alpha + \frac{\psi}{\phi_p q + \phi_w(1-q)}(q\phi_p(u_p(t_p^L) - u_p(t_p^R)) + (1-q)\phi_w(u_w(t_w^L) - u_w(t_w^R)))$$

Therefore, party $L$'s goal is to select a taxation platform that maximizes its own probability of victory, taking into consideration party $R$'s tax platform. Party $L$'s problem is then

$$\begin{aligned} \max \quad & P\left(s^L(\mathbf{t}^L, \mathbf{t}^R) > \frac{1}{2}\right) \\ \text{subject to} \quad & t_p^L q y_p + t_w^L(1-q)y_w = D \\ & t_p^L, t_w^L \leq 1 \\ & t_p^L, t_w^L \geq 0 \end{aligned}$$

Party $R$'s problem can be defined analogously. In fact, $R$ and $L$ face the same optimization problem. The equilibrium of this game is characterized by feasible tax platforms $\mathbf{t}^{L\star}$ and $\mathbf{t}^{R\star}$ such that neither party can improve its chances of victory by switching to another platform.

## A.2 Results

The primary quantity of interest in the model is progressiveness' of the equilibrium tax platforms, defined as follows.

**Definition 2 (Tax progressiveness)** *The progressiveness of a tax platform $(t_p, t_w)$ is given by the difference in the tax rate paid by the rich and the tax rate paid by the poor, $t_w - t_p$. If $t_w - t_p > 0$, we say the tax platform is progressive. If $t_w - t_p < 0$, we say the tax platform is regressive.*

Furthermore, when comparing tax platforms $(t_p, t_w)$ and $(t'_p, t'_w)$, I say platform $(t_p, t_w)$ is "more progressive" than platform $(t'_p, t'_w)$ if the difference $t_w - t_p$ is greater than $t'_w - t'_p$.

I now describe the equilibrium and comparative statics of the model. First, the political parties will converge on the same tax platform in equilibrium, a result of the fact that both political parties have the same preferences over policy and therefore solve the same optimization problem.

**Proposition 1** *In equilibrium, both parties propose identical tax platforms.*

$$t_p^{L\star} = t_p^{R\star} \qquad\qquad t_w^{L\star} = t_w^{R\star}$$

Henceforth, I will refer to the equilibrium policies as $(t_p^\star, t_w^\star)$, doing away with the party superscript.

Second, the expansion of government debt increases the tax burden of both social classes. However, the degree to which this additional debt is borne primarily by the wealthy or the poor is conditioned by economic inequality and the sensitivity to tax policy between the classes. In particular, when the ratio of income between the wealthy and poor ($\frac{y_w}{y_p}$) sufficiently exceeds the ratio of ideological homogeneity ($\frac{\phi_w}{\phi_p}$), the burden of additional debt is placed primarily on the wealthy. The statement of this requirement is given in Proposition 2. This behavior occurs because the political parties face a trade-off when setting tax policy: raising a group's taxes helps pay off debt but drives away voters. Political candidates consequently tax the social group that offers a better rate of tax revenue to lost votes. This process is isomorphic to the "electoral rates of return" dynamic described in the model of redistributive politics model by Cox and McCubbins (1986). Rather than distributing more benefits to groups that offer the greatest electoral advantages as in the model of Cox and McCubbins (1986), the parties in this model instead impose costs on the groups that are least likely to switch their vote. It follows that, in highly unequal societies, greater debt, such as that accumulated through war, can drive tax progressiveness.[14]

**Proposition 2** *In equilibrium, greater debt obligations increase tax progressiveness under the following condition*

$$\frac{y_w}{y_p} > \frac{\phi_w}{\phi_p}\frac{u_w''(t_w^\star)}{u_p''(t_p^\star)} \implies \frac{\partial[t_w^\star - t_p^\star]}{\partial D} > 0$$

The model also demonstrates that war fatalities, or any other variable that creates greater class consciousness, leads to more progressive tax policy in equilibrium. This operates through the same "rate of return" mechanism as the result on debt. Because an increase war fatalities ($\uparrow \alpha$) also produces an increase in class consciousness among the poor ($\uparrow \phi_p$), the poor become more sensitive to taxation. As a result, the electoral cost of a marginal increase in the tax rate on the poor also rises, leading parties to shift more of the tax burden onto the wealthy.

**Proposition 3** *In equilibrium, an increase in war fatalities ($\uparrow \alpha$) induces the candidates to propose more progressive tax platforms.*

$$\frac{\partial[t_w^\star - t_p^\star]}{\partial \alpha} > 0$$

Finally, as might be expected, an increase in war fatalities ($\uparrow \alpha$) also increases the expected share of the vote received by $L$, the left-wing party. This follows from the fact that war fatalities increase the size of the "fairness" parameter generating ideological support for party $L$.

**Proposition 4** *In equilibrium, an increase in war fatalities ($\uparrow \alpha$) increases the expected share of the vote received by party $L$.*

$$\frac{\partial E_\delta[s^L(\mathbf{t}^{L\star}, \mathbf{t}^{R\star})]}{\partial \alpha} > 0$$

[14] This of course assumes that the wealthy cannot use their financial position to lobby for lower taxes.

The equilibrium of the model highlights how changes in government debt can confound an observed correlation between mass mobilization and progressive taxation. This indicates that it may be problematic to ascribe the emergence progressive taxation following a war to ideological shifts about the proper distribution of the tax burden between social classes. Progressive tax policy may have instead resulted from electoral competition between class-based political parties over how to pay for the increased debt that fighting a war imposes.

However, Propositions 3 and 4 suggest an empirical approach that may get around this issue. A debt increase due to war is a national-level shock, but there is constituency-level variation in exposure to mass warfare at the local level in the form of war fatalities. If such exposure does influence individual beliefs about who should pay for the war, particularly in the form of greater ideological homogeneity, we should see candidates of both parties proposing more progressive tax platforms in constituencies with greater exposure to war.

## A.3 Proposition 1

**Proposition 1** *In equilibrium, both parties propose identical tax platforms.*

$$t_p^{L\star} = t_p^{R\star} \qquad\qquad t_w^{L\star} = t_w^{R\star}$$

**Proof.** Party $L$'s maximization problem is

$$\begin{aligned} \max \;\; & P\left(s^L(\mathbf{t}^L, \mathbf{t}^R) > \frac{1}{2}\right) \\ \text{subject to} \;\; & t_p^L q y_p + t_w^L (1-q) y_w = D \\ & t_p^L, t_w^L \leq 1 \\ & t_p^L, t_w^L \geq 0 \end{aligned}$$

Substituting $t_w^L = \frac{D - q t_p^L y_p}{(1-q) y_w}$ and taking the derivative gives the first order optimality condition

$$\frac{\psi}{q\phi_p + (1-q)\phi_w}\left[q\phi_p u_p'(t_p^L) - (1-q)\phi_w u_w'\left(\frac{D - q t_p^L y_p}{(1-q)y_w}\right)\left(\frac{q y_p}{(1-q) y_w}\right)\right] = 0 \tag{5}$$

Likewise, party $R$'s maximization problem is

$$\begin{aligned} \max \;\; & 1 - P\left(s^L(\mathbf{t}^L, \mathbf{t}^R) > \frac{1}{2}\right) \\ \text{subject to} \;\; & t_p^R q y_p + t_w^R (1-q) y_w = D \\ & t_p^R, t_w^R \leq 1 \\ & t_p^R, t_w^R \geq 0 \end{aligned}$$

Following the same procedure as for party $L$ gives the first-order optimality condition

$$\frac{\psi}{q\phi_p + (1-q)\phi_w}\left[q\phi_p u_p'(t_p^R) - (1-q)\phi_w u_w'\left(\frac{D - q t_p^R y_p}{(1-q)y_w}\right)\left(\frac{q y_p}{(1-q) y_w}\right)\right] = 0$$

Note that the parties' first-order optimality conditions are identical and that the platform of the opposing party is not present. Consequently, the optimization problems are identical and have the same solution. Taking the second derivative verifies that the objective function is concave and that the solution will be an equilibrium.

$$\frac{\psi}{q\phi_p + (1-q)\phi_w}\left[q\phi_p u_p''(t_p^R) + (1-q)\phi_w u_w''\left(\frac{D - qt_p^R y_p}{(1-q)y_w}\right)\left(\frac{qy_p}{(1-q)y_w}\right)^2\right] < 0$$

where the inequality follows from the concavity of the utility functions $u_g$ and the fact that all other exogenous parameters are strictly positive. ■

## A.4 Proposition 2

**Proposition 2** *In equilibrium, greater debt obligations increase tax progressiveness under the following condition*

$$\frac{y_w}{y_p} > \frac{\phi_w}{\phi_p}\frac{u_w''(t_w^\star)}{u_p''(t_p^\star)} \implies \frac{\partial[t_w^\star - t_p^\star]}{\partial D} > 0$$

**Proof.** Treating the equilibrium tax rate $t_P^\star$ as a function of the debt level $D$ and differentiating equation 5 with respect to $D$ produces

$$\frac{\psi}{q\phi_p + (1-q)\phi_w}\left[q\phi_p u_p''(t_p^\star)\frac{\partial t_p^\star}{\partial D} - (1-q)\phi_w u_w''\left(\frac{D - qt_p^\star y_p}{(1-q)y_w}\right)\left(\frac{qy_p}{(1-q)y_w}\right)\left(\frac{1 - q\frac{\partial t_p^\star}{\partial D}y_p}{(1-q)y_w}\right)\right] = 0$$

Re-arranging and isolating $\frac{\partial t_p^\star}{\partial D}$ on the left-hand side yields

$$\frac{\partial t_p^\star}{\partial D} = \frac{(1-q)\phi_w u_w''\left(\frac{D-qt_p^\star y_p}{(1-q)y_w}\right)\frac{qy_p}{(1-q)^2 y_w^2}}{q\phi_p u_p''(t_p^\star) + (1-q)\phi_w u_w''\left(\frac{D-qt_p^\star y_p}{(1-q)y_w}\right)\frac{q^2 y_p^2}{(1-q)^2 y_w^2}} > 0$$

where the inequality follows because the numerator and both terms in the denominator are negative, a consequence of the concavity of $u_g$. Likewise, the derivative of the tax on the wealthy $t_w^\star$ with respect to $D$ is

$$\frac{\partial t_w^\star}{\partial D} = \frac{\partial}{\partial D}\left[\frac{D - qt_p^\star y_p}{(1-q)y_w}\right] = \frac{1 - q\frac{\partial t_p^\star}{\partial D}y_p}{(1-q)y_w} > 0$$

The equilibrium tax progressiveness will be increasing in $D$ if

$$\frac{\partial[t_w^\star - t_p^\star]}{\partial D} = \frac{1 - q\frac{\partial t_p^\star}{\partial D}y_p}{(1-q)y_w} - \frac{\partial t_p^\star}{\partial D} > 0 \iff$$

$$\frac{1}{(1-q)y_w} - \frac{\partial t_p^\star}{\partial D}\left(1 + \frac{qy_p}{(1-q)y_w}\right) > 0 \iff$$

$$\left(1 + \frac{qy_p}{(1-q)y_w}\right)\left(\frac{(1-q)\phi_w u_w''\left(\frac{D-qt_p^\star y_p}{(1-q)y_w}\right)\frac{qy_p}{(1-q)^2 y_w^2}}{q\phi_p u_p''(t_p^\star) + (1-q)\phi_w u_w''\left(\frac{D-qt_p^\star y_p}{(1-q)y_w}\right)\frac{q^2 y_p^2}{(1-q)^2 y_w^2}}\right) < \frac{1}{(1-q)y_w}$$

Some algebra and simplifying shows the condition for the derivative to be positive to be

$$\frac{y_w}{y_p} > \frac{\phi_w}{\phi_p} \frac{u_w''(t_w^\star)}{u_p''(t_p^\star)}$$

■

## A.5 Lemma 1

This lemma is used in the proof of Proposition 3.

**Lemma 1** *In equilibrium, an increase in ideological homogeneity among the poor ($\uparrow \phi_p$) induces the candidates to propose more progressive tax platforms.*

$$\frac{\partial [t_w^\star - t_p^\star]}{\partial \phi_p} > 0$$

**Proof.** As before, I treat the equilibrium tax rate $t_P^\star$ as a function of $\phi_p$ and differentiate equation 5 with respect to $\phi_p$ to obtain

$$\frac{-q}{q\phi_p + (1-q)\phi_w} \underbrace{\frac{\partial P\left(s^L(\mathbf{t}^L, \mathbf{t}^R) > \frac{1}{2}\right)}{\partial t_p^L}}_{=0} + \frac{\psi}{q\phi_p + (1-q)\phi_w} \left( q u_p'(t_p^\star) + q\phi_p u_p''(t_p^\star) \frac{\partial t_p^\star}{\partial \phi_p} + (1-q)\phi_w u_w'' \left( \frac{D - q t_p^\star y_p}{(1-q) y_w} \right) \frac{q^2 y_p^2}{(1-q)^2 y_w^2} \frac{\partial t_p^\star}{\partial \phi_p} \right) = 0$$

Note that the term on the left is simply the first-order optimality condition times a constant, which must be 0 in equilibrium. Re-arranging then produces

$$\frac{\partial t_p^\star}{\partial \phi_p} = \frac{-q u_p'(t_p^\star)}{q\phi_p u_p''(t_p^\star) + (1-q)\phi_w u_w'' \left( \frac{D - q t_p^\star y_p}{(1-q) y_w} \right) \frac{q^2 y_p^2}{(1-q)^2 y_w^2}} < 0$$

where the inequality follows because the numerator is positive, due to the fact that $u_g$ is decreasing, while the denominator is negative, due to the concavity of $u_g$.

The result prior shows that the equilibrium tax rate on the poor decreases in $\phi_p$. This means that the tax rate on the wealthy must therefore increase in $\phi_p$, as verified below.

$$\frac{\partial t_w^\star}{\partial \phi_p} = -\frac{q y_p}{(1-q) y_w} \frac{\partial t_p^\star}{\partial \phi_p} > 0$$

Taken together, this means that tax progressiveness in equilibrium is increasing in $\phi_p$ ■

## A.6 Proposition 3

**Proposition 3** *In equilibrium, an increase in war fatalities ($\uparrow \alpha$) induces the candidates to propose more progressive tax platforms.*

$$\frac{\partial [t_w^\star - t_p^\star]}{\partial \alpha} > 0$$

**Proof.** Take the derivative of tax progressiveness with respect to $\alpha$. By the chain rule

$$\frac{\partial[t_w^\star - t_p^\star]}{\partial\alpha} = \frac{\partial[t_w^\star - t_p^\star]}{\partial\phi_p}\frac{\partial\phi_p}{\partial\alpha} > 0$$

The inequality follows from Lemma 1 and the fact that $\phi_p$ is an increasing function of $\alpha$ (which means $\frac{\partial\phi_p}{\partial\alpha}$ is positive). ■

## A.7 Proposition 4

**Proposition 4** *In equilibrium, an increase in war fatalities ($\uparrow \alpha$) increases the expected share of the vote received by party $L$.*

$$\frac{\partial E_\delta[s^L(\mathbf{t}^{L\star}, \mathbf{t}^{R\star})]}{\partial\alpha} > 0$$

**Proof.** Note that because the parties choose the same tax platform in equilibrium, the share of the vote received by party $L$ is

$$s^L(\mathbf{t}^{L\star}, \mathbf{t}^{R\star}) = \frac{1}{2} - q\phi_p\delta(\alpha) - (1-q)\phi_w\delta(\alpha)$$

Taking the expectation over $\delta(\alpha)$ yields

$$E_\delta[s^L(\mathbf{t}^{L\star}, \mathbf{t}^{R\star})] = \frac{1}{2} + q\phi_p\alpha + (1-q)\phi_w\alpha$$

Differentiating with respect to $\alpha$

$$\frac{\partial E_\delta[s^L(\mathbf{t}^{L\star}, \mathbf{t}^{R\star})]}{\partial\alpha} = q\frac{\partial\phi_p}{\partial\alpha}\alpha + q\phi_p + (1-q)\phi_w > 0$$

■

# B Data appendix

## B.1 Sample fatality record

Below is a sample record from the Commonwealth War Graves Commission database of fallen World War I soldiers. I extract soldiers' home addresses from the "Additional Info" box, outlined in red in Figure A.1.

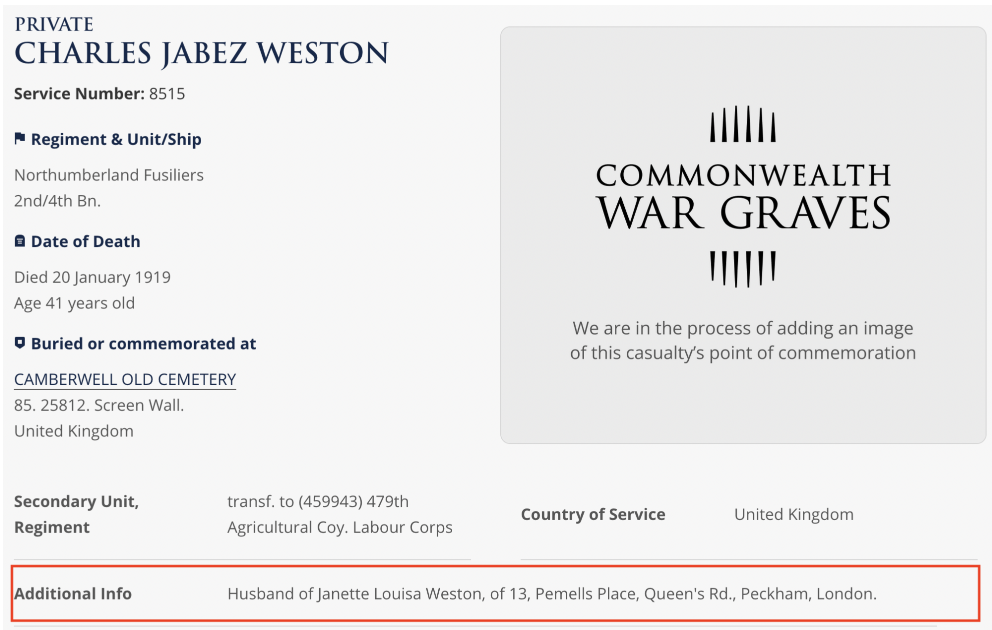


Figure A.1: Sample fatality record

## B.2 LLM prompt

Below is the text used to prompt ChatGPT to extract soldier addresses.

> You will be provided with a series of short epitaphs. Some epitaphs contain an address. From each epitaph, extract only the address or state 'None' if there is no address. If there are addresses for both a wife and parents in the text, extract the one that corresponding to the wife. Do not include any other text in your response.

Figure A.2: ChatGPT Prompt

## B.3 Geocoding covariate balance

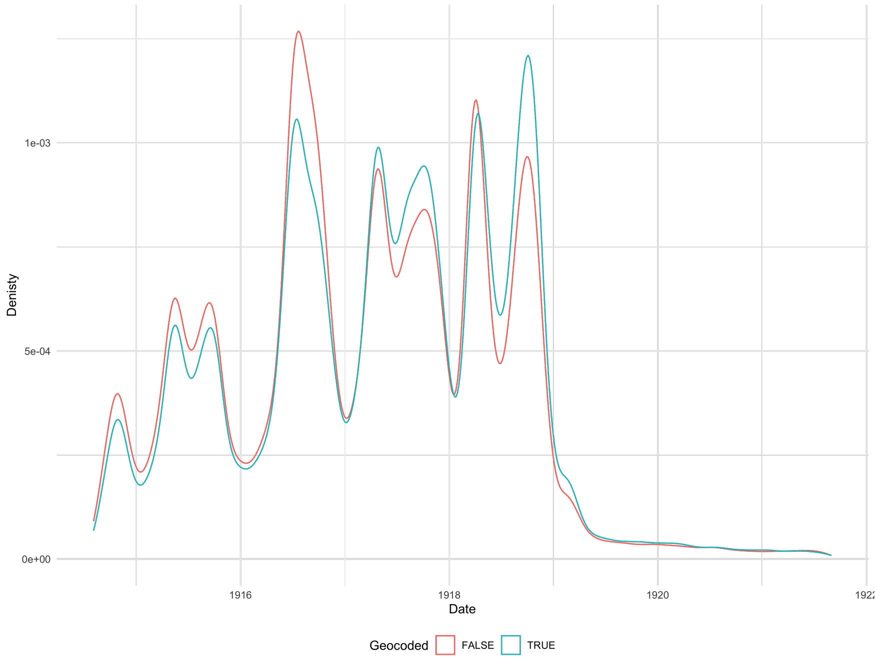


Figure A.3: Balance by date of death

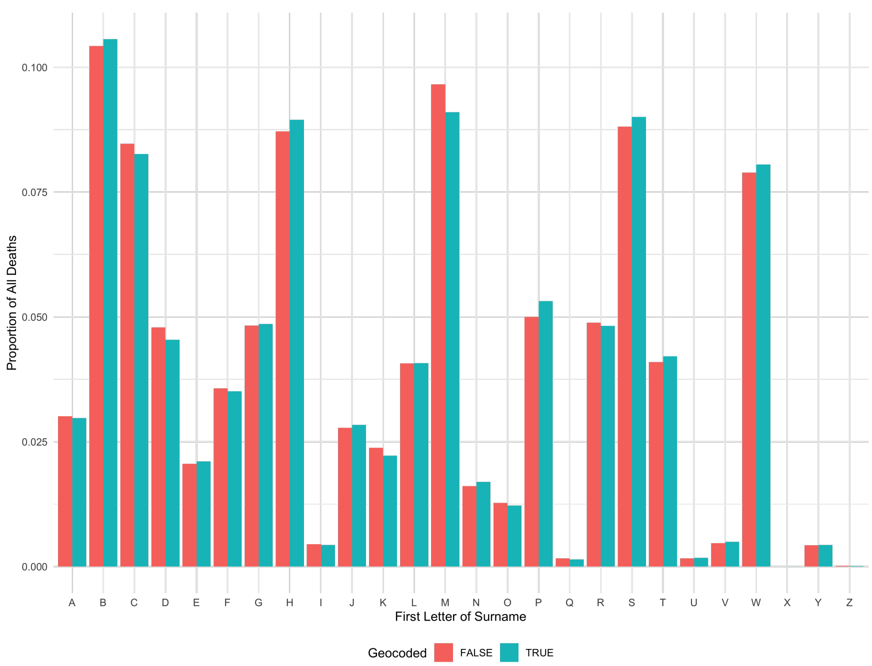


Figure A.4: Balance by first letter of surname

Figures A.3 and A.4 plot the distributions of geocoded and non-geocoded fatality records by date of death and first letter of surname. In both cases the distributions appear similar. Table A.1

| Variable | Geocoded | Non-geocoded |
|---|---|---|
| Average Age | 26.36 | 28.39 |
| Percent Private | 56 | 61.2 |

Table A.1: Balance by age and rank

plots the average age and proportion of soldiers holding the rank of private for geocoded and non-geocoded fatality records. The average values are close, though geocoded soldiers tended to be somewhat younger and less likely to hold the rank of private, indicating that higher-ranked officers may have had better records kept after their death. It is worth nothing that non-geocoded soldiers had far more missing data: only 60,882 records of 377,944 (16%) had a recorded value for age at death, though this includes records that had no address information whatsoever. By comparison, 419,216 of 445,260 (94%) geocoded records had age information availabile.

## B.4 List of left-wing parties

Table A.2 lists UK parties I code as being economically left-wing that also had candidates contest a House of Commons constituency in general elections between 1918 and 1935.

| **Party/Affiliation** |
|---|
| British Labour Party |
| British Socialist Party |
| Co-operative Party |
| Communist Party of Great Britain |
| Highland Land League |
| Independent Labour Party |
| Independent communist/progressive/socialist |
| National Democratic and Labour Party |
| National Socialist Party |
| Socialist Labour Party |

Table A.2: UK left-wing parties, 1918-1935

## B.5 Tax legislation coding rules

The table below describes under what conditions “Aye” and “No” votes are considered to indicate support for progressive taxation for each category of taxation.

| **Topic** | **Aye** | **No** |
|---|---|---|
| Capital levy | The proposed legislation imposes a capital levy or nationalizes property; or it takes steps to do so | The proposed legislation limits the ability of the government to nationalize property or impose a capital levy |
| Customs duty | The proposed legislation reduces the rate of or abolishes an existing tariff | The proposed legislation imposes a new tariff or increases the rate of an existing tariff |
| Excise duty | The proposed legislation reduces the rate of or abolishes an existing non-tariff consumption tax | The proposed legislation imposes a new non-tariff consumption tax or increases the rate of an existing non-tariff consumption tax |
| Income tax | The proposed legislation introduces new tax or modifies existing tax that increases gradation of the income tax (i.e., creates more tax brackets, raises taxes on the rich or lowers them on the poor) | The proposed legislation introduces new tax or modifies existing tax that reduces gradation of the income tax (i.e., collapses tax brackets, lowers taxes on the rich or increases them on the poor) |
| Wealth taxes | The proposed legislation imposes a new tax on capital (i.e., property, securities), increases the rate of an existing tax on capital, or reduces exemptions/increases enforcement of a tax on capital | The proposed legislation abolishes or reduces the rate of an existing tax capital; or it increases exemptions and/or reduces enforcement of a tax on capital |

Table A.3: Coding rules for tax legislation

### B.6 Example legislation

The following is a clause proposed by Mr. Trevelyan Thomson, MP for Middlesbrough West, on June 21, 1922. The clause seeks to increase the gradation of the income tax by adding two additional tax brackets above the base level.

> Section twenty-three of the Finance Act 1920 (which provides for a reduced rate of Income Tax on the first two hundred and twenty-five pounds of taxable income) shall have effect as if there were added the following:The rate on which the next two hundred pounds of the taxable income of an individual shall be charged to Income Tax shall he two-thirds of the standard rate of tax, and the rate on which the next following two hundred pounds of the taxable income of an individual shall be charged to Income Tax shall be five-sixths of the standard rate of tax.

The question moved, i.e., the formal statement that MPs must cast a yes-or-no vote on, is "That the Clause be read a Second time." In Westminster-style Parliamentary debate, a second reading of a piece of legislation opens the legislation to debate, allowing other MPs to amend it. If the legislation does not obtain a second reading, it is effectively killed. Thus, because a "No" vote aids the defeat of legislation that would increase the gradation of income tax, an "Aye" vote is coded as being in favor of progressive taxation,

### B.7 Descriptive statistics

| Variable | N | Mean | Std. Dev. | Min | Pctl. 25 | Pctl. 75 | Max |
|---|---|---|---|---|---|---|---|
| Fatality rate | 509 | 6.9 | 3 | 1.3 | 5 | 8.1 | 27 |
| Population | 509 | 70814 | 21486 | 16111 | 58676 | 78514 | 188835 |
| Population density | 509 | 4846 | 7631 | 23 | 155 | 6429 | 54592 |
| Share employed in agriculture | 509 | 0.063 | 0.072 | 0.0052 | 0.012 | 0.1 | 0.28 |
| Share employed in manual labor | 509 | 0.86 | 0.048 | 0.68 | 0.83 | 0.9 | 0.94 |
| Educators per capita | 509 | 0.0072 | 0.0022 | 0.00061 | 0.0058 | 0.0086 | 0.015 |
| Share women | 509 | 0.52 | 0.025 | 0.41 | 0.5 | 0.53 | 0.62 |
| Unemployed or indigent | 509 | 211 | 204 | 14 | 82 | 257 | 2126 |
| Average household size | 509 | 4.5 | 0.31 | 3.8 | 4.3 | 4.7 | 6.3 |
| Average age | 509 | 28 | 1.9 | 23 | 27 | 30 | 33 |
| Area | 509 | 303 | 462 | 1.2 | 11 | 461 | 3119 |
| Distance from large city | 509 | 23 | 28 | 0 | 3.3 | 34 | 172 |

Table A.4: Constituency descriptive statistics

## B.8 Prewar covariate balance

This appendix details the procedure used to conduct the balance tests described in Section 3.3 and provides the results in table form. For each prewar demographic/political variable, I estimate the following equation using ordinary least squares

$$fatalityRate_i = \alpha_c + \beta x_i + \epsilon_{ic}$$

where $fatalityRate_i$ is the conscript fatality rate of constituency $i$, $\alpha_c$ is an administrative county fixed effect, $x_i$ is a constituency-level predictor of interest, and $\epsilon_{ic}$ is an error term. The unit of analysis is a House of Commons parliamentary constituency, using boundaries as they were between 1885 and 1918. Standard errors are clustered at the level of the administrative county. The results are displayed in Table A.5. All reported coefficients are standardized.

| | DV: Fatality rate (%) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 | Model 6 | Model 7 | Model 8 | Model 9 | Model 10 | Model 11 | Model 12 |
| Conservative vote share (%) | 0.08 | | | | | | | | | | | |
| | (0.10) | | | | | | | | | | | |
| LW/Liberal vote share (%) | | −0.07 | | | | | | | | | | |
| | | (0.08) | | | | | | | | | | |
| Turnout (%) | | | 0.03 | | | | | | | | | |
| | | | (0.07) | | | | | | | | | |
| Votes for People's Budget | | | | −0.05 | | | | | | | | |
| | | | | (0.06) | | | | | | | | |
| Average age | | | | | 0.16 | | | | | | | |
| | | | | | (0.10) | | | | | | | |
| Average household size | | | | | | −0.01 | | | | | | |
| | | | | | | (0.05) | | | | | | |
| Log(population density) | | | | | | | $-0.15^{*}$ | | | | | |
| | | | | | | | (0.09) | | | | | |
| Share in farming (%) | | | | | | | | 0.03 | | | | |
| | | | | | | | | (0.07) | | | | |
| Share in manual labor (%) | | | | | | | | | −0.06 | | | |
| | | | | | | | | | (0.10) | | | |
| Share in professions (%) | | | | | | | | | | 0.06 | | |
| | | | | | | | | | | (0.10) | | |
| Share unemployed (%) | | | | | | | | | | | 0.03 | |
| | | | | | | | | | | | (0.06) | |
| Share woman (%) | | | | | | | | | | | | 0.08 |
| | | | | | | | | | | | | (0.06) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Num. clusters: | 58 | 62 | 59 | 62 | 62 | 62 | 62 | 62 | 62 | 62 | 62 | 62 |
| N | 352 | 468 | 389 | 463 | 468 | 468 | 468 | 468 | 468 | 468 | 468 | 468 |
| $R^2$ | 0.22 | 0.21 | 0.19 | 0.21 | 0.22 | 0.21 | 0.22 | 0.21 | 0.21 | 0.21 | 0.21 | 0.21 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.5: Table for pre-war balance

# C Additional results and robustness checks

## C.1 Additional election outcome specifications

Table A.6 displays additional specifications used to evaluate the effect of war fatalities on support for left-wing candidates with pooled (combining all election cycles) data. Column 1 reports the simple bivariate regression of left-wing vote share on the World War I combat fatality rate. Column 2 adds to this prewar demographic and geographic control variables. Columns 3 and 4 introduce administrative county and election fixed effects. Column 5 includes county $\times$ election fixed effects to account for time-varying regional unobservables. Finally, Column 6 features constituency-level linear trends as an alternative method to account for time-varying unobservables. In all models, the magnitude, direction, and statistical significance of the coefficient on the fatality rate remain similar to the baseline model presented in Section 4.2.

| | DV: Left-wing vote share (%) | | | | | |
|---|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 | Model 6 |
| Fatality rate (%) | $0.38^{*}$ | $0.49^{***}$ | $0.37^{**}$ | $0.39^{**}$ | $0.36^{**}$ | $0.45^{**}$ |
| | (0.20) | (0.13) | (0.15) | (0.15) | (0.16) | (0.22) |
| Avg. left-wing vote (%) | 37.32 | 36.79 | 36.79 | 36.79 | 36.79 | 36.79 |
| Avg. margin of victory (%) | 21.02 | 21.02 | 21.02 | 21.02 | 21.02 | 21.02 |
| Controls | - | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | - | - | ✓ | ✓ | ✓ | ✓ |
| Election FEs | - | - | - | ✓ | ✓ | ✓ |
| County $\times$ Election FEs | - | - | - | - | ✓ | - |
| Constituency trend | - | - | - | - | - | ✓ |
| Num. clusters: | 503 | 493 | 493 | 493 | 493 | 493 |
| N | 2886 | 2832 | 2832 | 2832 | 2832 | 2832 |
| $R^2$ | 0.01 | 0.52 | 0.59 | 0.65 | 0.70 | 0.84 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by constituency.

Table A.6: More pooled models on effect of WW1 fatalities on left-wing vote share

## C.2 Additional MP voting specifications

Table A.7 displays additional specifications used to evaluate the effect of war fatalities on MP support for progressive taxation legislation with pooled (combining all parliaments) data. Column 1 report the estimate with a party fixed effect alone. Columns 2 and 3 add year and county fixed effects, respectively. Column 4 includes constituency-level control variables and Column 5 introduces a party $\times$ year fixed effect. Standard errors are clustered by constituency to account for autocorrelation in the outcome variable. In all models, the magnitude, direction, and statistical significance of the coefficient on the fatality rate remain similar to the baseline model presented in Section 4.4.

| | DV: Num. pro-progressive tax votes | | | | |
|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 |
| Fatality rate (%) | 0.05** | 0.05** | 0.04** | 0.07*** | 0.06** |
| | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) |
| Controls | - | - | - | ✓ | ✓ |
| Party FEs | ✓ | ✓ | ✓ | ✓ | - |
| Year FEs | - | ✓ | ✓ | ✓ | - |
| County FEs | - | - | ✓ | ✓ | ✓ |
| Party $\times$ Year FEs | - | - | - | - | ✓ |
| DV Mean: | 4.67 | 4.69 | 4.69 | 4.69 | 4.69 |
| Num. clusters: | 508 | 508 | 508 | 498 | 498 |
| N | 9045 | 9045 | 9045 | 8866 | 8866 |
| $R^2$ | 0.49 | 0.62 | 0.63 | 0.63 | 0.84 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by constituency.

Table A.7: Effect of WW1 fatalities on MP support for progressive taxation

## C.3 WWI fatalities and electoral candidates

To test whether local World War I fatalities influenced the number of candidates contesting a House of Commons seat, I estimate the following equation for each general election cycle using ordinary least squares

$$candidates_{ice} = \alpha_c + \beta fatalityRate_i + \delta^T \mathbf{X}_i + \epsilon_{ice}$$

where $candidates_{ice}$ is the number of candidates contesting constituency $i$ in county $c$ during election $e$, $\alpha_c$ is a county fixed effect, $\mathbf{X}_i$ is a vector of prewar geographic and demographic controls, $\epsilon_{ice}$ is an error term, and $fatalityRate_i$ is the percentage of conscripted soldiers from constituency $i$ that died during the First World War. I include the same set of controls used in the analyses of left-wing vote share and parliamentary voting, as they contain variables that may influence how competitive a seat is. The full table for results in this section can be found in Appendix D.3.1.

| | DV: Number of candidates | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality Rate (%) | 0.02 | 0.00 | 0.03 | 0.01 | 0.02** | 0.01 | 0.01 | 0.02 |
| | (0.02) | (0.02) | (0.02) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. num. of candidates | 2.35 | 2.42 | 2.43 | 2.40 | 2.91 | 2.17 | 2.27 | 2.42 |
| Num. clusters: | 58 | 58 | 58 | 58 | 58 | 58 | 58 | 58 |
| N | 499 | 497 | 498 | 498 | 498 | 498 | 498 | 3486 |
| $R^2$ | 0.23 | 0.27 | 0.31 | 0.27 | 0.23 | 0.22 | 0.26 | 0.13 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.8: Effect of WW1 fatalities on number of candidates

Results obtained from estimating these equations are displayed in Table A.8. Reported standard errors are clustered at the level of the administrative county. In every general election except that of 1929, a constituency's WWI fatalities do not have a statistically significant effect on the number of candidates choosing to contest that constituency's House of Commons seat. Moreover, in all models, the size of the effect is tiny. Even in 1929, a one percentage point increase in a constituency's fatality rate is associated with an only a 0.02 increase in the number of candidates contesting an election, compared to a mean value of 2.91 candidates. Moving from the minimum fatality rate (1.3%) to the maximum (26.9%) is associated with a 0.56 increase in the number of candidates. The small size of these effects and their general lack of statistical significance provides strong evidence that WWI fatalities do not have an effect on the number of candidates contesting a constituency. Consequently, I treat the number of candidates as similar to a pre-treatment control that does not bias estimates of the effect of WWI fatalities on the vote share received by left-wing candidates.

## C.4 Conditioning on 1910 left-wing vote share

In this section, I reanalyze the effect of war fatalities on left-wing vote share while conditioning for 1910 left-wing vote share rather than 1910 Conservative vote share. Note that because Labour and other economically leftist parties were less prominent before the war, this approach limits the conclusions to those constituencies that left-wing parties regarded as competitive enough to be worth contesting. The overall results are similar, while the election-level results have larger point estimates in most cases, suggesting that war fatalities had a greater effect in constituencies where Labor was already established prior to the war. The full table for results in this section can be found in Appendix D.3.2.

| | DV: Left-wing vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality Rate (%) | 0.62*** | 0.45* | 1.11*** | 0.35 | 0.16 | 0.12 | 0.39* | 0.35** |
| | (0.22) | (0.23) | (0.28) | (0.28) | (0.20) | (0.17) | (0.21) | (0.15) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. left-wing share (%) | 31.92 | 38.30 | 40.19 | 37.35 | 38.69 | 33.09 | 40.83 | 37.32 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 22 | 24 | 22 | 25 | 25 | 25 | 27 | 27 |
| N | 172 | 193 | 189 | 205 | 233 | 217 | 219 | 1428 |
| $R^2$ | 0.45 | 0.69 | 0.69 | 0.71 | 0.74 | 0.74 | 0.72 | 0.53 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.9: Effect of WW1 fatalities on left-wing vote share (1910 left-wing control)

## C.5 Left-wing electoral margin

In this section, I test whether the results linking WWI fatalities to left-wing support are robust to an alternative measure of left-wing electoral success: margin of victory/loss. I define the left-wing margin as difference in constituency vote share received by a left-wing candidate and either the runner-up candidate or the victor (if a non-left-wing candidate won).

$$\text{Left-wing margin} = \begin{cases} \frac{\text{Left-wing votes}-\text{Runner-up votes}}{\text{Turnout}} \times 100 & \text{Left-wing winner} \\ \frac{\text{Left-wing votes}-\text{Winner votes}}{\text{Turnout}} \times 100 & \text{Non-left-wing winner} \end{cases}$$

To examine how war fatalities influenced left-wing margin, I estimate the following via OLS

$$leftwingMargin_{ice} = \alpha fatalityRate_i + \gamma^T \mathbf{X}_i + \delta_c + \nu_e + \epsilon_{ice}$$

where $leftwingMargin_{ice}$ is the electoral margin, as defined above, of the left-wing candidate(s) in constituency $i$ in county $c$ during election $e$. I include the same set of controls used in the analyses of left-wing vote share and parliamentary voting, save for the number of candidates. The full table for results in this section can be found in Appendix D.3.3. The estimated effects of war fatalities are similar in size and significance to those on vote share.

| | DV: Left-wing electoral margin (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality Rate (%) | 0.62 | 0.55 | 1.03** | 0.53 | 1.15** | 0.31 | 0.52 | 0.68** |
| | (0.37) | (0.34) | (0.41) | (0.39) | (0.45) | (0.28) | (0.33) | (0.27) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. left-wing margin (%) | −24.49 | −9.62 | −3.35 | −14.58 | −0.98 | −29.85 | −13.53 | −13.52 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 47 | 54 | 51 | 54 | 57 | 53 | 56 | 58 |
| N | 327 | 354 | 363 | 422 | 481 | 434 | 451 | 2832 |
| $R^2$ | 0.42 | 0.61 | 0.56 | 0.70 | 0.67 | 0.75 | 0.73 | 0.49 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.10: Effect of WW1 fatalities on left-wing electoral margin

## C.6 Conservative vote share

To examine the effect of local war fatalities on Conservative vote share, I estimate the following equation using ordinary least squares

$$conShare_{ice} = \alpha fatalityRate_i + \beta candidates_{ice} + \gamma^T \mathbf{X}_i + \delta_c + \nu_e + \epsilon_{ice}$$

where $conShare_{ice}$ is the share of the vote received by a Conservative Party candidate contesting constituency $i$ in county $c$ during election $e$, with the rest of the specification identical to that of Equation 1 for left-wing share. The results are reported in Table A.11. Columns 1 through 7 display the estimated effect of war fatalities on Conservative vote share for each election cycle, while Column 8 reports the estimated effect for the entire interwar period. Full tables for results in this section can be found in Appendix D.3.4. In both cases, there is no evidence that war fatalities increased or decreased the share of the vote received by Conservatives.

| | DV: Conservative vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality Rate (%) | −0.12 | −0.07 | 0.02 | 0.18 | 0.09 | 0.31 | 0.08 | 0.12 |
| | (0.18) | (0.21) | (0.21) | (0.19) | (0.22) | (0.29) | (0.24) | (0.16) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. Con. vote share (%) | 57.46 | 48.93 | 43.14 | 52.22 | 39.19 | 62.88 | 54.41 | 50.62 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 47 | 51 | 53 | 54 | 58 | 50 | 52 | 58 |
| N | 307 | 381 | 423 | 428 | 482 | 393 | 409 | 2823 |
| $R^2$ | 0.61 | 0.64 | 0.59 | 0.68 | 0.73 | 0.72 | 0.71 | 0.54 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.11: Effect of WW1 fatalities on Conservative vote share

## C.7 Liberal vote share

To examine the effect of local war fatalities on Liberal vote share, I estimate the following equation using ordinary least squares

$$libShare_{ice} = \alpha fatalityRate_i + \beta candidates_{ie} + \gamma^T \mathbf{X}_i + \delta_c + \nu_e + \epsilon_{ice}$$

where $libShare_{ice}$ is the share of the vote received by a Liberal Party candidate contesting constituency $i$ in county $c$ during election $e$, with the rest of the specification identical to that of Equation 1 for left-wing share. The results are reported in Table A.12. Columns 1 through 7 display the estimated effect of war fatalities on Liberal vote share for each election cycle, while Column 8 reports the estimated effect for the entire interwar period. Full tables for results in this section can be found in Appendix D.3.5. The pooled results show there is no statistically detectable effect of fatalities on Liberal vote share. However, this appears to be due to significant heterogeneity in the effect over time.

| | DV: Liberal vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality Rate (%) | 0.47 | 0.13 | 0.27 | $-0.17$ | $-0.22$ | $-1.76^{**}$ | $-1.27^{**}$ | 0.09 |
| | (0.34) | (0.21) | (0.30) | (0.40) | (0.21) | (0.78) | (0.60) | (0.21) |
| Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. Liberal vote share (%) | 27.57 | 32.69 | 37.74 | 30.95 | 27.77 | 36.68 | 25.42 | 31.29 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 45 | 50 | 54 | 55 | 58 | 41 | 46 | 58 |
| N | 236 | 272 | 372 | 289 | 443 | 98 | 143 | 1853 |
| $R^2$ | 0.53 | 0.64 | 0.58 | 0.63 | 0.48 | 0.74 | 0.77 | 0.43 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.12: Effect of WW1 fatalities on Liberal vote share

# D Full tables

## D.1 Electoral outcomes

This appendix contains the full version of Table 2.

| | DV: Left-wing vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | 0.36** | 0.34** | 0.66*** | 0.32 | 0.43** | 0.15 | 0.33** | 0.37*** |
| | (0.16) | (0.16) | (0.23) | (0.24) | (0.18) | (0.13) | (0.16) | (0.13) |
| 1910 Conservative vote share (%) | −0.13** | −0.12*** | −0.12* | −0.11*** | −0.09** | −0.12*** | −0.14*** | −0.11*** |
| | (0.05) | (0.04) | (0.06) | (0.04) | (0.04) | (0.04) | (0.05) | (0.03) |
| Number of candidates | −4.30*** | −8.06*** | −11.46*** | −8.93*** | −5.67*** | −3.44*** | −5.73*** | −6.05*** |
| | (1.08) | (0.75) | (1.30) | (1.23) | (1.04) | (0.71) | (1.08) | (0.41) |
| Log(Pop. density) | −0.30 | 2.02*** | 1.03 | 1.63* | 1.48* | 0.94 | 0.17 | 1.00 |
| | (0.88) | (0.65) | (0.99) | (0.94) | (0.81) | (0.78) | (0.95) | (0.73) |
| Log(Population) | 1.71 | 3.62 | 9.94** | 1.01 | 1.33 | 0.84 | 0.98 | 2.05 |
| | (1.96) | (2.56) | (4.00) | (2.80) | (2.35) | (1.85) | (2.33) | (1.84) |
| Average age | −2.11** | −2.20*** | −1.98** | −3.60*** | −4.31*** | −3.66*** | −5.63*** | −3.69*** |
| | (0.88) | (0.80) | (0.87) | (0.48) | (0.48) | (0.60) | (0.41) | (0.40) |
| Average household size | −1.41 | −3.59 | 0.01 | −8.56*** | −9.94*** | −6.44** | −11.52*** | −6.53*** |
| | (3.10) | (2.91) | (2.75) | (2.11) | (2.25) | (2.56) | (2.11) | (1.98) |
| Share of woman (%) | −0.50 | −0.87** | −0.89** | −0.37 | −0.79** | −0.88** | −0.24 | −0.58** |
| | (0.39) | (0.40) | (0.37) | (0.31) | (0.34) | (0.42) | (0.40) | (0.28) |
| Educators per capita | 619.57 | 261.67 | −555.90 | −103.16 | 48.52 | 879.14** | 187.68 | 197.25 |
| | (864.16) | (808.30) | (1081.65) | (550.59) | (652.17) | (408.86) | (451.79) | (585.31) |
| Share unemployed or indigent (%) | 0.62 | 1.16 | −0.40 | 2.42*** | 1.81* | 2.39** | 3.25*** | 1.80** |
| | (0.65) | (1.32) | (1.26) | (0.77) | (1.08) | (1.08) | (0.81) | (0.73) |
| Share in manual labor (%) | 0.77** | 1.07*** | 0.35 | 0.94*** | 1.02*** | 1.27*** | 0.70*** | 0.91*** |
| | (0.32) | (0.27) | (0.47) | (0.27) | (0.22) | (0.18) | (0.26) | (0.21) |
| Share in agriculture (%) | −0.43 | −0.06 | −0.24 | −0.47** | −0.85*** | −0.99*** | −0.88*** | −0.59*** |
| | (0.31) | (0.16) | (0.31) | (0.23) | (0.22) | (0.20) | (0.24) | (0.19) |
| Log(Dist. from large city) | 0.69 | 0.06 | −0.10 | 0.25 | 0.31 | 0.51 | 2.04*** | 0.50 |
| | (0.71) | (0.75) | (0.93) | (0.72) | (0.70) | (0.51) | (0.62) | (0.57) |
| Area (km$^2$) | 0.00 | −0.00 | −0.00 | 0.00 | 0.00 | 0.00** | 0.00* | 0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. left-wing vote share (%) | 31.92 | 38.30 | 40.19 | 37.35 | 38.69 | 33.09 | 40.83 | 37.32 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 47 | 54 | 51 | 54 | 57 | 53 | 56 | 58 |
| N | 327 | 354 | 363 | 422 | 481 | 434 | 451 | 2832 |
| $R^2$ | 0.46 | 0.69 | 0.69 | 0.74 | 0.78 | 0.75 | 0.76 | 0.59 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.13: Full table for effect of WW1 fatalities on left-wing vote share

## D.2 Parliamentary voting

This appendix contains full versions of tables reported in Section 4.4.

| | DV: Num. pro-progressive tax votes | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1918-1922 | 1922-1923 | 1923-1924 | 1924-1929 | 1929-1931 | 1931-1935 | Pooled |
| Fatality rate (%) | $0.10^{**}$ | $0.12^{*}$ | 0.05 | 0.09 | $0.14^{**}$ | $-0.01$ | $0.07^{***}$ |
| | (0.05) | (0.06) | (0.03) | (0.07) | (0.06) | (0.02) | (0.02) |
| 1910 Conservative vote share (%) | $-0.00$ | 0.02 | 0.00 | $-0.01$ | $-0.01$ | $-0.01^{*}$ | $-0.00$ |
| | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.00) | (0.00) |
| Log(Pop. density) | $-0.00$ | 0.14 | $-0.04$ | $-0.23$ | 0.23 | $-0.11$ | $-0.04$ |
| | (0.15) | (0.27) | (0.22) | (0.30) | (0.22) | (0.09) | (0.11) |
| Log(Population) | 0.37 | $-0.24$ | $0.76^{**}$ | $-0.10$ | 1.11 | $-0.01$ | 0.21 |
| | (0.43) | (0.58) | (0.31) | (0.44) | (0.75) | (0.18) | (0.16) |
| Average age | $-0.09$ | $-0.48^{**}$ | 0.10 | $-0.36^{*}$ | $-0.15$ | $-0.06$ | $-0.12$ |
| | (0.11) | (0.19) | (0.15) | (0.21) | (0.16) | (0.07) | (0.07) |
| Average household size | $-0.07$ | $-1.52^{**}$ | 0.13 | $-1.91^{***}$ | $-0.41$ | 0.29 | $-0.62^{**}$ |
| | (0.46) | (0.73) | (0.51) | (0.69) | (0.46) | (0.20) | (0.25) |
| Share of women (%) | $-0.10$ | $0.21^{*}$ | $-0.05$ | 0.06 | 0.04 | 0.03 | 0.03 |
| | (0.07) | (0.11) | (0.11) | (0.13) | (0.08) | (0.04) | (0.06) |
| Educators per capita | 142.52 | $-88.95$ | $-91.95$ | $-149.62$ | 11.71 | $-79.30$ | $-64.41$ |
| | (98.70) | (137.95) | (83.61) | (135.63) | (115.85) | (49.96) | (79.97) |
| Share unemployed or indigent (%) | 0.23 | 0.72 | 0.00 | 0.23 | 0.25 | 0.00 | 0.04 |
| | (0.29) | (0.45) | (0.23) | (0.24) | (0.23) | (0.13) | (0.14) |
| Share in manual labor (%) | $0.12^{**}$ | 0.00 | $-0.04$ | 0.07 | 0.01 | $-0.01$ | 0.03 |
| | (0.05) | (0.07) | (0.05) | (0.09) | (0.07) | (0.02) | (0.04) |
| Share in agriculture (%) | $-0.05$ | 0.05 | 0.00 | 0.02 | 0.03 | $-0.00$ | 0.01 |
| | (0.05) | (0.06) | (0.06) | (0.05) | (0.04) | (0.02) | (0.03) |
| Log(Dist. from large city) | $-0.10$ | 0.17 | 0.02 | $-0.01$ | $0.29^{*}$ | 0.04 | 0.03 |
| | (0.11) | (0.22) | (0.14) | (0.22) | (0.17) | (0.06) | (0.08) |
| Area (km$^2$) | $-0.00$ | $-0.00$ | 0.00 | $-0.00$ | 0.00 | $-0.00$ | $-0.00$ |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Party FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| DV Mean: | 3.47 | 7.31 | 5.24 | 7.58 | 6.36 | 1.92 | 4.69 |
| Num. clusters: | 57 | 58 | 58 | 58 | 58 | 58 | 58 |
| N | 1887 | 493 | 499 | 1974 | 1460 | 2344 | 8866 |
| R$^2$ | 0.68 | 0.84 | 0.61 | 0.74 | 0.71 | 0.68 | 0.63 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.14: Full table for effect of WW1 fatalities on MP support for progressive taxation

| | DV: Num. pro-progressive tax votes | | |
|---|---|---|---|
| | Conservative | Labour | Liberal |
| Fatality Rate (%) | 0.03*** | 0.10 | 0.02 |
| | (0.01) | (0.07) | (0.29) |
| 1910 Conservative vote share (%) | −0.00 | −0.00 | −0.05 |
| | (0.00) | (0.01) | (0.06) |
| Log(Pop. density) | 0.05* | −0.22 | −0.05 |
| | (0.02) | (0.35) | (1.19) |
| Log(Population) | 0.21*** | 0.80 | −1.31 |
| | (0.07) | (0.77) | (1.99) |
| Average age | −0.01 | −0.93** | −0.71 |
| | (0.01) | (0.34) | (0.74) |
| Average household size | −0.01 | −4.49** | −1.15 |
| | (0.08) | (1.71) | (3.93) |
| Share of women (%) | −0.01 | 0.12 | −0.17 |
| | (0.01) | (0.22) | (0.66) |
| Educators per capita | −0.06 | 116.38 | −299.45 |
| | (9.96) | (273.07) | (651.63) |
| Share unemployed or indigent (%) | −0.06** | 0.94* | 0.87 |
| | (0.03) | (0.53) | (1.91) |
| Share in manual labor (%) | 0.00 | −0.02 | −0.10 |
| | (0.01) | (0.16) | (0.37) |
| Share in agriculture (%) | −0.00 | 0.15 | −0.06 |
| | (0.01) | (0.16) | (0.28) |
| Log(Dist. from large city) | 0.04* | 0.10 | −0.27 |
| | (0.03) | (0.31) | (0.51) |
| Area (km$^2$) | 0.00 | −0.00 | 0.00 |
| | (0.00) | (0.00) | (0.01) |
| County × Year FEs | ✓ | ✓ | ✓ |
| DV Mean: | 0.67 | 15.07 | 8.68 |
| No. clusters: | 52 | 37 | 51 |
| N | 4492 | 1918 | 649 |
| $R^2$ | 0.64 | 0.80 | 0.84 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.15: Full table for effect of WW1 fatalities on MP support for progressive taxation by party

| | DV: Num. pro-progressive tax votes | | | | |
|---|---|---|---|---|---|
| | Income tax | Wealth tax | Capital levy | Excise duty | Customs duty |
| Fatality Rate (%) | 0.01 | $0.04^{***}$ | 0.00 | 0.01 | $0.02^{*}$ |
| | (0.01) | (0.01) | (0.00) | (0.01) | (0.01) |
| 1910 Conservative vote share (%) | $-0.00$ | $-0.00$ | $-0.00$ | $-0.00$ | $-0.00$ |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| Log(Pop. density) | 0.02 | 0.08 | 0.00 | 0.03 | $-0.05$ |
| | (0.04) | (0.07) | (0.01) | (0.02) | (0.08) |
| Log(Population) | $-0.02$ | 0.34 | $-0.02$ | 0.07 | $-0.02$ |
| | (0.09) | (0.21) | (0.02) | (0.05) | (0.13) |
| Average age | $-0.06^{**}$ | $-0.05$ | $-0.02^{**}$ | $-0.02$ | $-0.10$ |
| | (0.02) | (0.03) | (0.01) | (0.02) | (0.06) |
| Average household size | $-0.24^{**}$ | $-0.47^{***}$ | $-0.01$ | $-0.11$ | $-0.26$ |
| | (0.11) | (0.14) | (0.03) | (0.08) | (0.18) |
| Share of women (%) | 0.01 | $-0.03^{*}$ | 0.00 | 0.00 | 0.04 |
| | (0.02) | (0.02) | (0.01) | (0.01) | (0.04) |
| Educators per capita | 9.85 | $-29.66$ | 3.00 | 7.36 | $-42.81$ |
| | (24.61) | (32.79) | (6.25) | (15.93) | (77.31) |
| Share unemployed or indigent (%) | 0.05 | $0.19^{**}$ | 0.02 | $-0.01$ | $-0.04$ |
| | (0.08) | (0.09) | (0.02) | (0.03) | (0.11) |
| Share in manual labor (%) | 0.01 | $-0.02$ | $0.01^{*}$ | $0.01^{*}$ | $0.05^{*}$ |
| | (0.02) | (0.02) | (0.00) | (0.01) | (0.03) |
| Share in agriculture (%) | 0.00 | 0.01 | 0.00 | $-0.00$ | 0.01 |
| | (0.01) | (0.01) | (0.00) | (0.01) | (0.02) |
| Log(Dist. from large city) | 0.01 | $0.09^{***}$ | 0.00 | 0.00 | $-0.00$ |
| | (0.03) | (0.03) | (0.01) | (0.02) | (0.05) |
| Area (km$^2$) | $-0.00$ | $-0.00$ | $-0.00$ | 0.00 | $-0.00$ |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County × Year FEs | ✓ | ✓ | ✓ | ✓ | ✓ |
| Party FEs | ✓ | ✓ | ✓ | ✓ | ✓ |
| DV Mean: | 1.40 | 1.94 | 0.33 | 0.98 | 2.13 |
| Num. clusters: | 58 | 58 | 58 | 58 | 58 |
| N | 7040 | 4150 | 2064 | 5565 | 8253 |
| $R^2$ | 0.70 | 0.70 | 0.85 | 0.57 | 0.64 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.16: Full table for effect of WW1 fatalities on MP support for progressive taxation by topic

## D.3 Additional results

This appendix contains full versions of tables reported in Sections C.3, C.4, C.5, C.6, and C.7

### D.3.1 WWI fatalities and electoral candidates

| | DV: Number of candidates | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | 0.02 | 0.00 | 0.03 | 0.01 | $0.02^{**}$ | 0.01 | 0.01 | 0.02 |
| | (0.02) | (0.02) | (0.02) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) |
| 1910 Conservative vote share (%) | −0.00 | 0.00 | 0.00 | 0.00 | −0.00 | 0.00 | 0.00 | 0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| Log(Pop. density) | −0.03 | 0.00 | $0.11^{**}$ | 0.00 | 0.07 | 0.03 | 0.03 | 0.03 |
| | (0.07) | (0.07) | (0.05) | (0.06) | (0.04) | (0.07) | (0.05) | (0.04) |
| Log(Population) | 0.12 | $0.46^{***}$ | 0.39 | 0.33 | $0.48^{**}$ | $0.33^{**}$ | $0.42^{**}$ | $0.36^{**}$ |
| | (0.17) | (0.16) | (0.24) | (0.25) | (0.19) | (0.16) | (0.20) | (0.17) |
| Average age | $-0.12^{*}$ | 0.00 | −0.02 | −0.02 | −0.01 | −0.01 | $0.08^{**}$ | −0.02 |
| | (0.07) | (0.03) | (0.04) | (0.03) | (0.04) | (0.06) | (0.04) | (0.04) |
| Average household size | $-0.91^{***}$ | $-0.68^{***}$ | $-0.99^{***}$ | $-0.99^{***}$ | $-0.43^{***}$ | $-0.37^{**}$ | $-0.37^{***}$ | $-0.68^{***}$ |
| | (0.23) | (0.16) | (0.16) | (0.14) | (0.13) | (0.16) | (0.11) | (0.12) |
| Share of women (%) | 0.00 | −0.00 | $-0.05^{*}$ | $-0.03^{*}$ | −0.02 | −0.01 | −0.00 | −0.02 |
| | (0.04) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) |
| Educators per capita | $-90.41^{**}$ | −61.14 | −25.74 | −25.60 | 13.75 | −53.53 | $-44.31^{*}$ | $-40.66^{**}$ |
| | (36.61) | (37.77) | (32.82) | (29.05) | (23.22) | (38.91) | (24.27) | (19.21) |
| Share unemployed or indigent (%) | 0.03 | −0.05 | 0.03 | $0.15^{*}$ | 0.03 | $0.11^{*}$ | 0.04 | 0.05 |
| | (0.11) | (0.07) | (0.13) | (0.08) | (0.07) | (0.07) | (0.05) | (0.05) |
| Share in manual labor (%) | $-0.04^{*}$ | −0.01 | −0.02 | $-0.03^{*}$ | −0.00 | −0.01 | $-0.02^{*}$ | −0.02 |
| | (0.02) | (0.02) | (0.02) | (0.02) | (0.01) | (0.02) | (0.01) | (0.01) |
| Share in agriculture (%) | −0.02 | −0.03 | −0.02 | −0.02 | $-0.02^{**}$ | $-0.03^{*}$ | −0.02 | $-0.02^{**}$ |
| | (0.02) | (0.02) | (0.01) | (0.02) | (0.01) | (0.02) | (0.02) | (0.01) |
| Log(Dist. from large city) | $0.16^{**}$ | $0.08^{**}$ | $0.16^{***}$ | $0.12^{***}$ | $0.12^{***}$ | 0.07 | 0.06 | $0.11^{***}$ |
| | (0.07) | (0.04) | (0.05) | (0.04) | (0.03) | (0.05) | (0.04) | (0.03) |
| Area (km$^2$) | −0.00 | −0.00 | −0.00 | −0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. num. of candidates | 235.32 | 241.75 | 242.69 | 240.00 | 290.76 | 217.14 | 226.55 | 241.88 |
| Num. clusters: | 58 | 58 | 58 | 58 | 58 | 58 | 58 | 58 |
| N | 499 | 497 | 498 | 498 | 498 | 498 | 498 | 3486 |
| $R^2$ | 0.23 | 0.27 | 0.31 | 0.27 | 0.23 | 0.22 | 0.26 | 0.13 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.17: Full table for effect of WW1 fatalities on number of candidates

### D.3.2 Conditioning on 1910 left-wing vote share

| | DV: Left-wing vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | $0.62^{***}$ | $0.45^{*}$ | $1.11^{***}$ | 0.35 | 0.16 | 0.12 | $0.39^{*}$ | $0.35^{**}$ |
| | (0.22) | (0.23) | (0.28) | (0.28) | (0.20) | (0.17) | (0.21) | (0.15) |
| 1910 left-wing vote share (%) | $0.19^{**}$ | $0.10^{**}$ | $0.08^{*}$ | $0.07^{*}$ | $0.09^{***}$ | $0.07^{**}$ | $0.11^{***}$ | $0.10^{**}$ |
| | (0.09) | (0.04) | (0.04) | (0.04) | (0.03) | (0.03) | (0.03) | (0.04) |
| Number of candidates | $-4.16^{***}$ | $-7.57^{***}$ | $-11.91^{***}$ | $-9.21^{***}$ | $-7.36^{***}$ | $-3.78^{***}$ | $-6.97^{***}$ | $-5.94^{***}$ |
| | (1.22) | (1.02) | (1.26) | (1.61) | (1.02) | (1.04) | (1.07) | (0.56) |
| Log(Pop. density) | −0.19 | 1.40 | 0.92 | 1.01 | 1.02 | 0.57 | −0.87 | 0.60 |
| | (1.34) | (0.84) | (1.20) | (0.95) | (0.89) | (0.90) | (1.30) | (0.79) |
| Log(Population) | 0.40 | $6.56^{**}$ | $15.54^{***}$ | $5.97^{***}$ | $4.44^{*}$ | $4.13^{*}$ | $5.41^{***}$ | $4.52^{**}$ |
| | (3.70) | (2.45) | (3.60) | (2.13) | (2.48) | (2.04) | (1.75) | (1.87) |
| Average age | −1.63 | −1.81 | 0.86 | −2.74 | $-4.82^{***}$ | $-2.90^{***}$ | $-6.21^{***}$ | $-3.06^{***}$ |
| | (2.19) | (1.72) | (1.50) | (1.70) | (1.22) | (0.82) | (1.04) | (1.09) |
| Average household size | −0.76 | −4.28 | 6.84 | −8.11 | $-12.94^{**}$ | −5.10 | $-16.42^{***}$ | −6.00 |
| | (6.91) | (5.76) | (6.88) | (6.88) | (5.01) | (3.42) | (4.09) | (4.23) |
| Share of women (%) | −0.11 | −0.80 | $-1.41^{**}$ | −0.53 | $-0.99^{*}$ | $-1.41^{**}$ | −0.10 | $-0.82^{*}$ |
| | (0.90) | (0.83) | (0.63) | (0.59) | (0.50) | (0.57) | (0.60) | (0.44) |
| Educators per capita | 41.83 | −516.57 | −1148.19 | −518.09 | −185.95 | 508.57 | −359.43 | −73.80 |
| | (969.19) | (829.84) | (750.40) | (530.59) | (590.26) | (493.83) | (676.59) | (447.41) |
| Share unemployed or indigent (%) | 0.02 | $2.21^{**}$ | −1.42 | 2.17 | $3.23^{**}$ | $2.40^{***}$ | $4.36^{***}$ | $1.87^{**}$ |
| | (0.91) | (0.88) | (2.37) | (1.31) | (1.26) | (0.65) | (0.71) | (0.68) |
| Share in manual labor (%) | 0.41 | $0.94^{**}$ | 0.12 | $0.72^{***}$ | $0.73^{***}$ | $1.19^{***}$ | 0.44 | $0.78^{***}$ |
| | (0.45) | (0.35) | (0.39) | (0.22) | (0.22) | (0.24) | (0.33) | (0.17) |
| Share in agriculture (%) | 0.31 | −0.02 | $-0.99^{*}$ | −0.48 | $-0.58^{*}$ | $-0.82^{***}$ | $-0.62^{*}$ | $-0.44^{*}$ |
| | (0.54) | (0.23) | (0.54) | (0.32) | (0.33) | (0.19) | (0.32) | (0.25) |
| Log(Dist. from large city) | $2.38^{*}$ | 0.31 | −0.06 | 0.46 | 0.98 | 0.14 | $1.62^{**}$ | 0.59 |
| | (1.21) | (1.32) | (1.06) | (0.89) | (1.09) | (0.44) | (0.69) | (0.81) |
| Area ($km^2$) | −0.01 | −0.00 | 0.00 | −0.00 | −0.00 | 0.00 | 0.00 | −0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. left-wing vote share (%) | 31.92 | 38.30 | 40.19 | 37.35 | 38.69 | 33.09 | 40.83 | 37.32 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 22 | 24 | 22 | 25 | 25 | 25 | 27 | 27 |
| N | 172 | 193 | 189 | 205 | 233 | 217 | 219 | 1428 |
| $R^2$ | 0.45 | 0.69 | 0.69 | 0.71 | 0.74 | 0.74 | 0.72 | 0.53 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.18: Full table for effect of WW1 fatalities on left-wing vote share (1910 left-wing)

### D.3.3 Left-wing electoral margin

| | DV: Left-wing electoral margin (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | 0.62 | 0.55 | $1.03^{**}$ | 0.53 | $1.15^{**}$ | 0.31 | 0.52 | $0.68^{**}$ |
| | (0.37) | (0.34) | (0.41) | (0.39) | (0.45) | (0.28) | (0.33) | (0.27) |
| 1910 Conservative vote share (%) | −0.18 | $-0.19^{**}$ | $-0.27^{**}$ | $-0.16^{**}$ | $-0.16^{**}$ | $-0.23^{***}$ | $-0.25^{***}$ | $-0.20^{***}$ |
| | (0.11) | (0.08) | (0.11) | (0.07) | (0.08) | (0.07) | (0.08) | (0.06) |
| Log(Pop. density) | −1.32 | $3.41^{*}$ | 0.85 | $2.87^{*}$ | $3.84^{**}$ | 2.38 | 0.26 | 2.09 |
| | (2.11) | (1.85) | (1.88) | (1.67) | (1.83) | (1.54) | (1.75) | (1.47) |
| Log(Population) | 5.54 | $8.59^{*}$ | $15.00^{**}$ | 0.62 | 11.84 | 3.23 | 1.01 | 6.11 |
| | (3.57) | (5.04) | (6.51) | (4.62) | (7.40) | (4.07) | (4.02) | (4.09) |
| Average age | $-3.81^{**}$ | $-4.08^{**}$ | $-3.48^{**}$ | $-6.64^{***}$ | $-7.51^{***}$ | $-7.77^{***}$ | $-10.74^{***}$ | $-6.86^{***}$ |
| | (1.59) | (1.53) | (1.71) | (1.04) | (1.04) | (1.20) | (0.86) | (0.76) |
| Average household size | −8.87 | $-11.15^{**}$ | 0.41 | $-17.23^{***}$ | $-24.33^{***}$ | $-17.40^{***}$ | $-23.20^{***}$ | $-17.04^{***}$ |
| | (5.61) | (4.98) | (5.00) | (3.63) | (4.20) | (3.76) | (4.02) | (3.25) |
| Share of women (%) | −0.97 | −1.13 | $-1.29^{**}$ | −0.50 | $-1.14^{*}$ | −1.15 | −0.27 | −0.87 |
| | (0.76) | (0.71) | (0.64) | (0.55) | (0.66) | (0.73) | (0.78) | (0.54) |
| Educators per capita | 192.45 | 254.96 | −891.69 | −285.85 | −28.72 | $1559.34^{*}$ | −115.84 | 130.64 |
| | (1330.00) | (1547.11) | (1777.05) | (1001.77) | (1346.14) | (849.14) | (864.41) | (1049.38) |
| Share unemployed or indigent (%) | 1.14 | 1.45 | −0.15 | $3.42^{***}$ | $3.37^{*}$ | $5.08^{**}$ | $5.99^{***}$ | $3.30^{***}$ |
| | (1.65) | (1.86) | (2.18) | (1.01) | (1.86) | (2.07) | (1.45) | (1.14) |
| Share in manual labor (%) | 0.99 | $2.07^{***}$ | 0.47 | $1.95^{***}$ | $1.57^{***}$ | $2.63^{***}$ | $1.29^{**}$ | $1.61^{***}$ |
| | (0.63) | (0.69) | (0.74) | (0.48) | (0.57) | (0.41) | (0.53) | (0.42) |
| Share in agriculture (%) | $-1.16^{*}$ | −0.28 | −0.82 | $-1.04^{***}$ | $-1.30^{***}$ | $-1.60^{***}$ | $-1.40^{***}$ | $-1.08^{***}$ |
| | (0.64) | (0.28) | (0.57) | (0.39) | (0.31) | (0.35) | (0.35) | (0.29) |
| Log(Dist. from large city) | $3.01^{**}$ | 0.60 | 0.24 | 1.54 | $2.92^{**}$ | $2.29^{**}$ | $4.40^{***}$ | $2.35^{**}$ |
| | (1.26) | (1.29) | (1.46) | (1.29) | (1.46) | (1.11) | (1.27) | (1.07) |
| Area ($km^2$) | 0.01 | −0.00 | −0.00 | 0.00 | 0.01 | $0.01^{*}$ | $0.01^{*}$ | $0.01^{*}$ |
| | (0.01) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. left-wing margin (%) | −24.49 | −9.62 | −3.35 | −14.58 | −0.98 | −29.85 | −13.53 | −13.52 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 47 | 54 | 51 | 54 | 57 | 53 | 56 | 58 |
| N | 327 | 354 | 363 | 422 | 481 | 434 | 451 | 2832 |
| $R^2$ | 0.42 | 0.61 | 0.56 | 0.70 | 0.67 | 0.75 | 0.73 | 0.49 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.19: Full table for effect of WW1 fatalities on left-wing electoral margin

### D.3.4 Conservative vote share

| | DV: Conservative vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | −0.12 | −0.07 | 0.02 | 0.18 | 0.09 | 0.31 | 0.08 | 0.12 |
| | (0.18) | (0.21) | (0.21) | (0.19) | (0.22) | (0.29) | (0.24) | (0.16) |
| 1910 Conservative vote share (%) | 0.00 | 0.08 | 0.05 | 0.06 | 0.08*** | 0.09*** | 0.10** | 0.07** |
| | (0.04) | (0.05) | (0.05) | (0.04) | (0.03) | (0.03) | (0.04) | (0.03) |
| Number of candidates | −8.45*** | −7.74*** | −4.51*** | −6.62*** | −3.42*** | −7.26*** | −6.90*** | −9.16*** |
| | (0.70) | (0.98) | (1.02) | (1.13) | (0.85) | (1.14) | (1.14) | (0.60) |
| Log(Pop. density) | 0.52 | −0.26 | 0.14 | −0.11 | −0.06 | 0.34 | −0.21 | 0.00 |
| | (1.17) | (0.86) | (0.80) | (0.76) | (0.49) | (0.98) | (0.74) | (0.60) |
| Log(Population) | −0.59 | 0.96 | −0.26 | 5.40*** | 2.79 | 6.42*** | 4.77** | 4.53*** |
| | (1.80) | (1.67) | (3.07) | (1.96) | (2.11) | (2.32) | (2.38) | (1.50) |
| Average age | 2.92*** | 1.97*** | 2.43** | 3.48*** | 4.89*** | 4.88*** | 5.39*** | 3.96*** |
| | (0.81) | (0.60) | (0.97) | (0.82) | (1.11) | (0.56) | (0.66) | (0.53) |
| Average household size | 8.06*** | 6.70** | 5.87* | 9.12*** | 15.21*** | 9.41*** | 9.87*** | 7.75*** |
| | (2.31) | (2.77) | (3.38) | (2.15) | (2.97) | (2.02) | (2.20) | (2.16) |
| Share of women (%) | −0.12 | −0.21 | −0.16 | −0.51* | −0.17 | −0.39 | 0.03 | −0.27 |
| | (0.46) | (0.24) | (0.25) | (0.29) | (0.32) | (0.35) | (0.40) | (0.24) |
| Educators per capita | 175.58 | 220.13 | 807.74* | 718.63* | 203.28 | 10.38 | 123.66 | 225.48 |
| | (749.62) | (435.47) | (409.27) | (373.77) | (263.94) | (420.82) | (385.60) | (251.99) |
| Share unemployed or indigent (%) | −0.95 | −1.11 | −0.19 | −1.82** | −2.92*** | −3.26*** | −3.02*** | −1.48*** |
| | (0.97) | (0.89) | (0.91) | (0.79) | (0.82) | (0.75) | (0.63) | (0.48) |
| Share in manual labor (%) | −0.78* | −0.88*** | −0.33* | −1.07*** | −0.87*** | −1.08*** | −0.73*** | −0.90*** |
| | (0.42) | (0.19) | (0.19) | (0.21) | (0.15) | (0.21) | (0.24) | (0.14) |
| Share in agriculture (%) | 0.01 | 0.18 | 0.32 | 0.27 | 0.34 | 0.39* | 0.47** | 0.15 |
| | (0.25) | (0.19) | (0.20) | (0.21) | (0.21) | (0.23) | (0.19) | (0.14) |
| Log(Dist. from large city) | −1.35* | −0.43 | −0.72 | −1.20*** | −1.21* | −1.55** | −2.16*** | −1.07** |
| | (0.79) | (0.44) | (0.57) | (0.44) | (0.64) | (0.67) | (0.78) | (0.41) |
| Area ($km^2$) | 0.00 | 0.00 | −0.00 | 0.00 | 0.00 | −0.00 | −0.00 | 0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. Con. vote share (%) | 57.46 | 48.93 | 43.14 | 52.22 | 39.19 | 62.88 | 54.41 | 50.62 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 47 | 51 | 53 | 54 | 58 | 50 | 52 | 58 |
| N | 307 | 381 | 423 | 428 | 482 | 393 | 409 | 2823 |
| $R^2$ | 0.61 | 0.64 | 0.59 | 0.68 | 0.73 | 0.72 | 0.71 | 0.54 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.20: Full table for effect of WW1 fatalities on Conservative vote share

### D.3.5 Liberal vote share

| | DV: Liberal vote share (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1918 | 1922 | 1923 | 1924 | 1929 | 1931 | 1935 | Pooled |
| Fatality rate (%) | 0.47 | 0.13 | 0.27 | −0.17 | −0.22 | −1.76** | −1.27** | 0.09 |
| | (0.34) | (0.21) | (0.30) | (0.40) | (0.21) | (0.78) | (0.60) | (0.21) |
| 1910 Conservative vote share (%) | −0.02 | 0.01 | 0.02 | 0.01 | 0.03 | 0.27** | 0.10 | 0.01 |
| | (0.06) | (0.04) | (0.05) | (0.05) | (0.03) | (0.13) | (0.06) | (0.03) |
| Number of candidates | −7.72*** | −11.79*** | −11.60*** | −15.11*** | −4.90*** | −14.40*** | −15.07*** | −11.29*** |
| | (1.27) | (0.97) | (1.27) | (1.65) | (1.10) | (2.00) | (2.81) | (0.65) |
| Log(Pop. density) | 0.29 | 1.26 | −0.25 | 0.43 | −1.43* | −1.33 | 1.05 | 0.26 |
| | (1.12) | (1.19) | (1.00) | (0.79) | (0.85) | (1.87) | (1.62) | (0.77) |
| Log(Population) | −1.90 | 0.42 | 5.76* | 5.99 | 0.12 | −12.13** | −5.98 | 1.82 |
| | (2.30) | (3.20) | (3.28) | (4.14) | (2.87) | (4.87) | (5.48) | (2.56) |
| Average age | −0.89 | −1.55** | −1.05 | −0.29 | −0.19 | 6.61 | −2.32 | −0.80 |
| | (1.20) | (0.75) | (0.93) | (1.27) | (1.04) | (4.39) | (1.53) | (0.64) |
| Average household size | −10.55** | −8.66*** | −9.25** | −1.87 | −2.26 | 26.27* | −4.67 | −5.26** |
| | (4.12) | (2.68) | (3.65) | (4.77) | (2.68) | (14.57) | (4.46) | (2.31) |
| Share of women (%) | −0.96* | 0.19 | 0.76** | 0.81 | 1.23*** | −0.18 | 1.48*** | 0.62** |
| | (0.57) | (0.43) | (0.36) | (0.55) | (0.40) | (1.52) | (0.51) | (0.25) |
| Educators per capita | −643.34 | −317.69 | −567.65 | −338.59 | −1479.95*** | −1780.07 | 680.24 | −627.51 |
| | (916.71) | (808.44) | (504.35) | (595.64) | (482.63) | (2252.69) | (1103.23) | (480.41) |
| Share unemployed or indigent (%) | −1.51 | −0.12 | 1.40 | −1.81 | −0.57 | 21.02*** | 3.72 | −0.20 |
| | (1.37) | (1.37) | (1.31) | (1.57) | (1.93) | (5.18) | (4.04) | (1.24) |
| Share in manual labor (%) | −0.36 | −0.19 | −0.32 | 0.29 | −0.39 | 2.16** | 0.95** | −0.04 |
| | (0.43) | (0.27) | (0.33) | (0.28) | (0.24) | (0.88) | (0.42) | (0.18) |
| Share in agriculture (%) | −0.14 | 0.56* | 0.03 | −0.07 | 0.64** | −1.05 | 0.46 | 0.30 |
| | (0.36) | (0.29) | (0.28) | (0.28) | (0.28) | (0.85) | (0.46) | (0.21) |
| Log(Dist. from large city) | 2.52** | 2.04*** | 0.66 | 0.85 | 1.51*** | −6.65** | 0.14 | 1.49*** |
| | (1.16) | (0.55) | (0.80) | (0.85) | (0.56) | (2.91) | (1.28) | (0.40) |
| Area ($km^2$) | 0.01 | 0.00 | 0.00 | 0.00 | −0.00** | 0.00 | 0.01 | 0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.01) | (0.01) | (0.00) |
| County FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Avg. Liberal vote share (%) | 27.57 | 32.69 | 37.74 | 30.95 | 27.77 | 36.68 | 25.42 | 31.29 |
| Avg. margin of victory (%) | 28.38 | 16.10 | 12.87 | 18.62 | 16.20 | 32.45 | 22.53 | 20.88 |
| Num. clusters: | 45 | 50 | 54 | 55 | 58 | 41 | 46 | 58 |
| N | 236 | 272 | 372 | 289 | 443 | 98 | 143 | 1853 |
| $R^2$ | 0.53 | 0.64 | 0.58 | 0.63 | 0.48 | 0.74 | 0.77 | 0.43 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by administrative county.

Table A.21: Full table for effect of WW1 fatalities on Liberal vote share

### D.3.6 Additional election outcome specifications

| | DV: Left-wing vote share (%) | | | | | |
|---|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 | Model 6 |
| Fatality rate (%) | 0.38* | 0.49*** | 0.37** | 0.39** | 0.36** | 0.45** |
| | (0.20) | (0.13) | (0.15) | (0.15) | (0.16) | (0.22) |
| 1910 Conservative vote share (%) | | −0.08*** | −0.11*** | −0.11*** | −0.11*** | −0.11*** |
| | | (0.03) | (0.03) | (0.03) | (0.03) | (0.04) |
| Number of candidates | | −6.70*** | −6.05*** | −7.17*** | −6.88*** | −6.68*** |
| | | (0.48) | (0.46) | (0.54) | (0.60) | (0.41) |
| Log(Pop. density) | | 1.22** | 1.00* | 0.96 | 0.89 | 0.69 |
| | | (0.59) | (0.60) | (0.60) | (0.64) | (0.92) |
| Log(Population) | | 1.74 | 2.05 | 2.57 | 2.34 | 3.67 |
| | | (1.55) | (1.67) | (1.73) | (1.85) | (2.31) |
| Average age | | −3.89*** | −3.69*** | −3.71*** | −3.71*** | −1.68* |
| | | (0.49) | (0.49) | (0.51) | (0.55) | (0.89) |
| Average household size | | −8.76*** | −6.53*** | −7.28*** | −7.09*** | −0.50 |
| | | (2.00) | (2.04) | (2.09) | (2.19) | (3.25) |
| Share of women (%) | | −0.44 | −0.58* | −0.58* | −0.61* | −0.72 |
| | | (0.33) | (0.31) | (0.32) | (0.34) | (0.45) |
| Educators per capita | | −481.59 | 197.25 | 121.18 | 140.85 | −97.68 |
| | | (380.86) | (383.12) | (380.56) | (403.86) | (629.94) |
| Share unemployed or indigent (%) | | 1.29* | 1.80*** | 1.87*** | 1.82** | −0.16 |
| | | (0.76) | (0.69) | (0.71) | (0.75) | (1.24) |
| Share in manual labor (%) | | 0.57*** | 0.91*** | 0.88*** | 0.89*** | 0.72** |
| | | (0.17) | (0.21) | (0.21) | (0.23) | (0.34) |
| Share in agriculture (%) | | −0.59*** | −0.59*** | −0.64*** | −0.59*** | −0.05 |
| | | (0.15) | (0.17) | (0.18) | (0.19) | (0.25) |
| Log(Dist. from large city) | | 1.27*** | 0.50 | 0.65 | 0.56 | −0.13 |
| | | (0.38) | (0.47) | (0.48) | (0.51) | (0.76) |
| Area ($km^2$) | | 0.00** | 0.00 | 0.00 | 0.00 | −0.00 |
| | | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| Avg. left-wing vote (%) | 37.32 | 36.79 | 36.79 | 36.79 | 36.79 | 36.79 |
| Avg. margin of victory (%) | 21.02 | 21.02 | 21.02 | 21.02 | 21.02 | 21.02 |
| County FEs | - | - | ✓ | ✓ | ✓ | ✓ |
| Election FEs | - | - | - | ✓ | ✓ | ✓ |
| County × Election FEs | - | - | - | - | ✓ | - |
| Constituency trend | - | - | - | - | - | ✓ |
| Num. clusters: | 503 | 493 | 493 | 493 | 493 | 493 |
| N | 2886 | 2832 | 2832 | 2832 | 2832 | 2832 |
| $R^2$ | 0.01 | 0.52 | 0.59 | 0.65 | 0.70 | 0.84 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by constituency.

Table A.22: Full table for more pooled models on effect of WW1 fatalities on left-wing vote share

### D.3.7 Additional parliamentary voting specifications

| | DV: Num. pro-progressive tax votes | | | | |
|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 |
| Fatality Rate (%) | $0.05^{**}$ | $0.05^{**}$ | $0.04^{**}$ | $0.07^{***}$ | $0.06^{**}$ |
| | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) |
| 1910 Conservative vote share (%) | | | | $-0.00$ | $-0.00$ |
| | | | | (0.00) | (0.00) |
| Log(Pop. density) | | | | $-0.04$ | $-0.03$ |
| | | | | (0.10) | (0.09) |
| Log(Population) | | | | 0.21 | 0.22 |
| | | | | (0.24) | (0.21) |
| Average age | | | | $-0.12$ | $-0.13^{**}$ |
| | | | | (0.08) | (0.07) |
| Average household size | | | | $-0.62^{**}$ | $-0.66^{***}$ |
| | | | | (0.29) | (0.23) |
| Share of women (%) | | | | 0.03 | 0.00 |
| | | | | (0.04) | (0.04) |
| Educators per capita | | | | $-64.41$ | $-41.53$ |
| | | | | (57.55) | (54.74) |
| Share unemployed or indigent (%) | | | | 0.04 | 0.17 |
| | | | | (0.17) | (0.13) |
| Share in manual labor (%) | | | | 0.03 | 0.02 |
| | | | | (0.03) | (0.03) |
| Share in agriculture (%) | | | | 0.01 | 0.01 |
| | | | | (0.02) | (0.02) |
| Log(Dist. from large city) | | | | 0.03 | 0.04 |
| | | | | (0.08) | (0.07) |
| Area (km$^2$) | | | | $-0.00$ | $-0.00$ |
| | | | | (0.00) | (0.00) |
| Party FEs | ✓ | ✓ | ✓ | ✓ | - |
| Year FEs | - | ✓ | ✓ | ✓ | - |
| County FEs | - | - | ✓ | ✓ | ✓ |
| Party × Year FEs | - | - | - | - | ✓ |
| DV Mean: | 4.67 | 4.69 | 4.69 | 4.69 | 4.69 |
| Num. clusters: | 508 | 508 | 508 | 498 | 498 |
| N | 9045 | 9045 | 9045 | 8866 | 8866 |
| $R^2$ | 0.49 | 0.62 | 0.63 | 0.63 | 0.84 |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Note: Robust SEs clustered by constituency.

Table A.23: Full table for effect of WW1 fatalities on MP support for progressive taxation